

Overset Grid Generation for Planetary Aerial Exploration

Patricia Ventura Diaz

NASA Ames Research Center

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patricia.venturadiaz@nasa.gov

From first flight to aerial exploration

Past and current NASA missions that use planetary aerial exploration







Unique **opportunities** and **challenges:**

- Vertical lift vehicles can hover and fly at low speeds, take-off and land at unprepared landing sites, perform panoramic survey, and have greater access to hazardous terrains.
- Planetary rotorcraft operate in regimes where the aerodynamics are unusual:
  - **Mars (Ingenuity + Skyfall)**: very low ρ (~1.5% Earth's), low Re, high M_{tip} → challenging conditions.
  - **Titan (Dragonfly)**: High ρ (4.4× Earth's) and low g ($1/7^{th}$ Earth's) → favorable conditions.

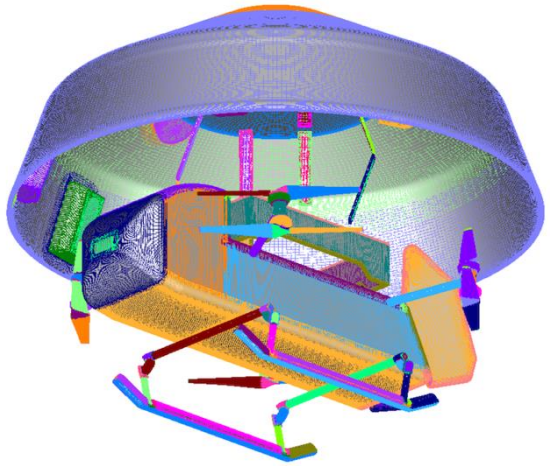
Complexity comes with a cost

More capable vehicles → more complex CFD configurations

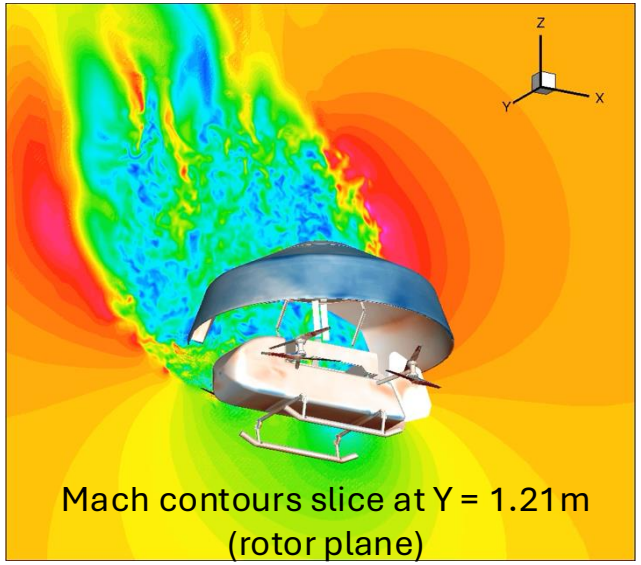
Dragonfly rotorcraft lander



Eight rotors with opposite cant.
Complex lander geometry..



Overset surface grids of Dragonfly lander + backshell

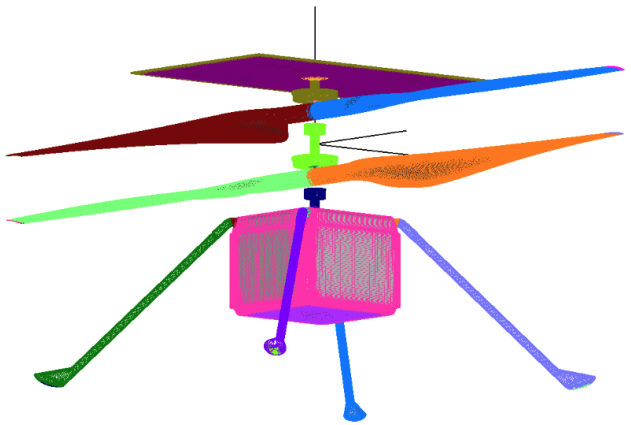


Mach contours slice at Y = 1.21m (rotor plane)

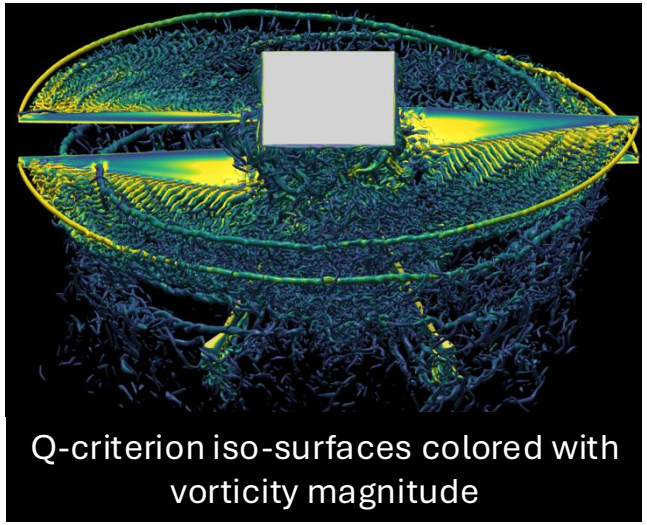
Skyfall helicopters



Two coaxial rotors.
Solar panel and main body.



Overset surface grids of Skyfall helicopter

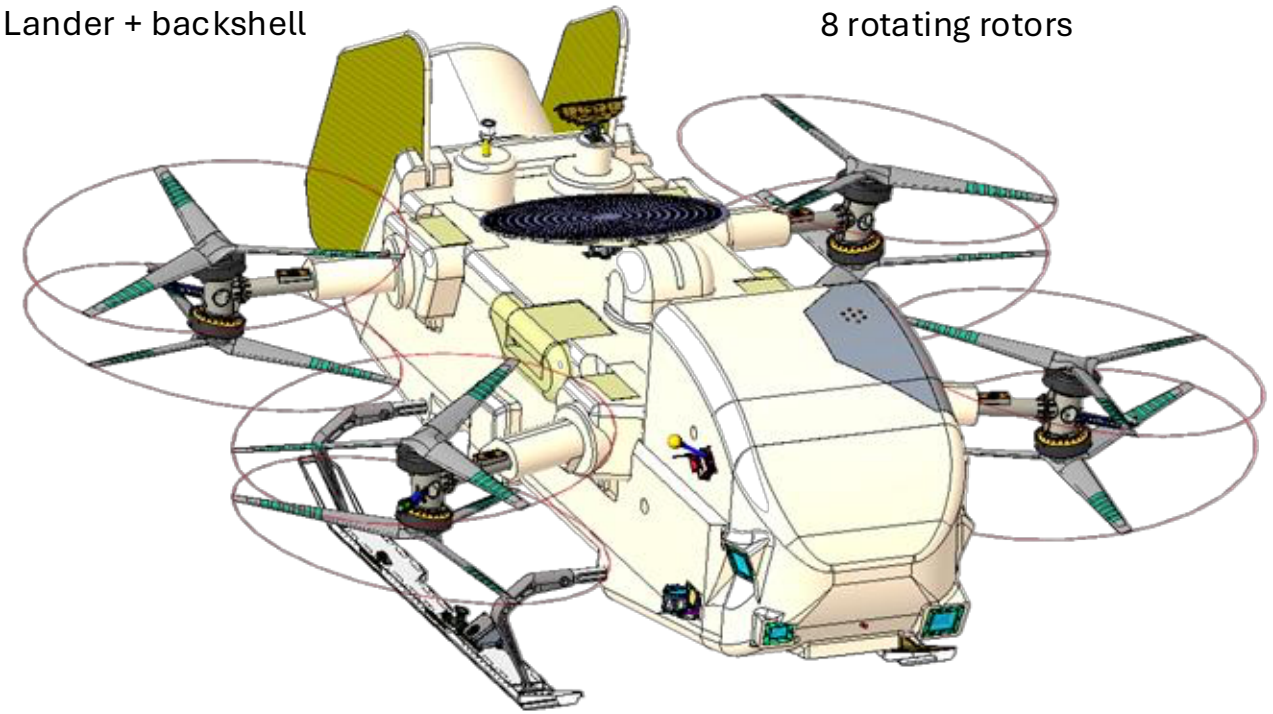


Q-criterion iso-surfaces colored with vorticity magnitude

Complex planetary vehicles require high-fidelity modeling

Complex geometry
Lander + backshell

Multiple moving components
8 rotating rotors



Aerodynamic interactions
Rotor–rotor / rotor–body

Unsteady wakes
Large range of relevant length scales

HIGH-FIDELITY CFD
- Blade-resolved
- Full 3D-unsteady flow
- Wake resolved

OVERSET GRIDS
- Component decomposition
- Relative motion
- Local refinement

Overset grids enable high-fidelity CFD of increasingly complex planetary rotorcraft configurations

OUTLINE

Numerical Approach

- High-Fidelity CFD solver: NASA's OVERFLOW
- Overset Grids: NASA's Chimera Grid Tools
- Practical example: ROAMX rotor + stand

Dragonfly Rotorcraft Lander

- Preparation for Powered Flight (PPF)
- NFAC PPF: OVERFLOW validation
- Titan PPF

Skyfall Helicopters

- Mission Overview
- Skyfall: overset grids
- Skyfall: hover solution

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High-fidelity CFD solver: NASA's OVERFLOW

Unsteady Reynolds-averaged Navier-Stokes **(URANS)**

Spatial Discretization:

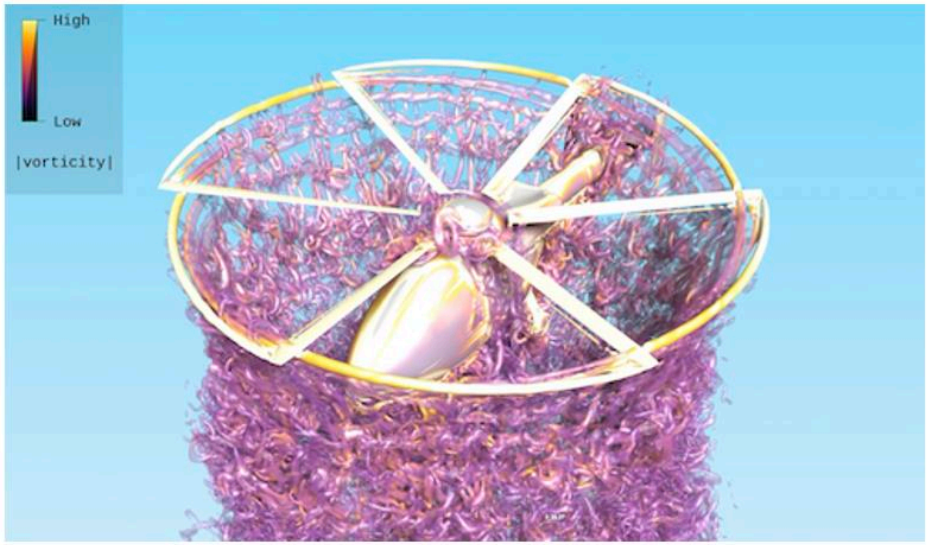
- Central/upwind differencing scheme
- 5th–order accuracy for convective terms
- 2nd–order accuracy for viscous terms

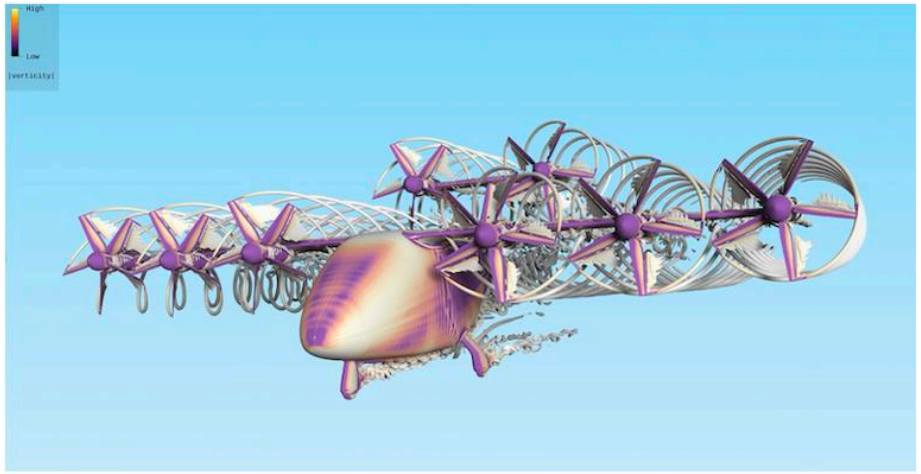
Temporal Integration:

- 2nd–order backward-difference (BDF2) implicit time integration
- Dual time stepping
- Angle step 0.25° per physical timestep

Hybrid RANS/LES turbulence model:

- SA 1-eq/SST 2-eq turbulence model
- DDES limiter
- y^{+}≤1

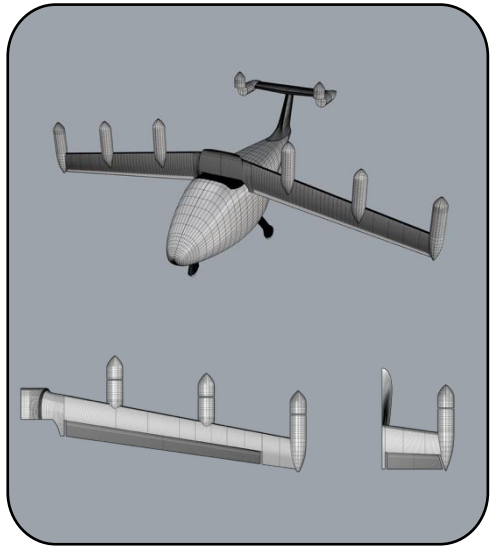




Overset Grids: NASA's Chimera Grid Tools (CGT)

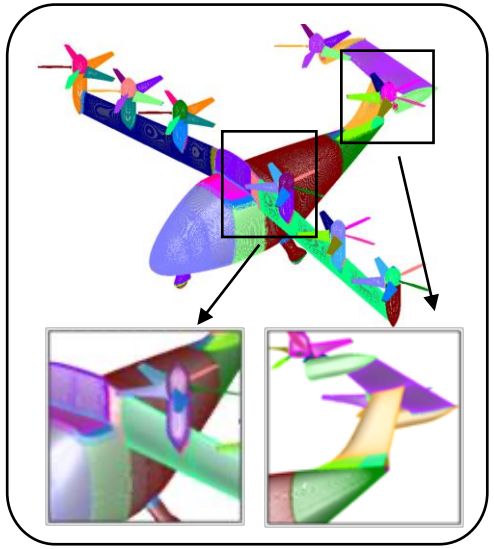
Geometry processing

- Clean geometry definition
- Smooth watertight surfaces from CAD file



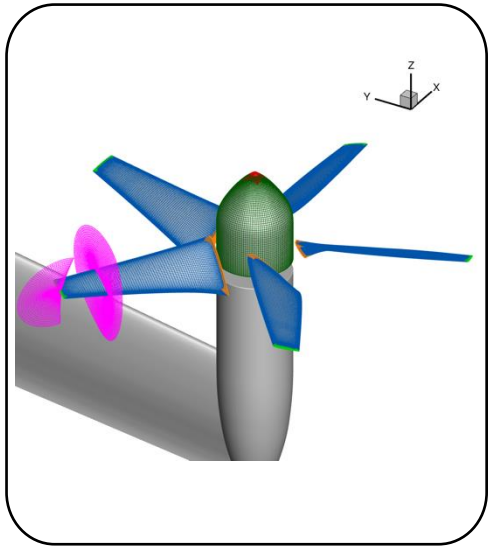
Surface grid generation

- Clustering in high-gradient regions
- Mesh generation with hyperbolic marching or transfinite interpolation



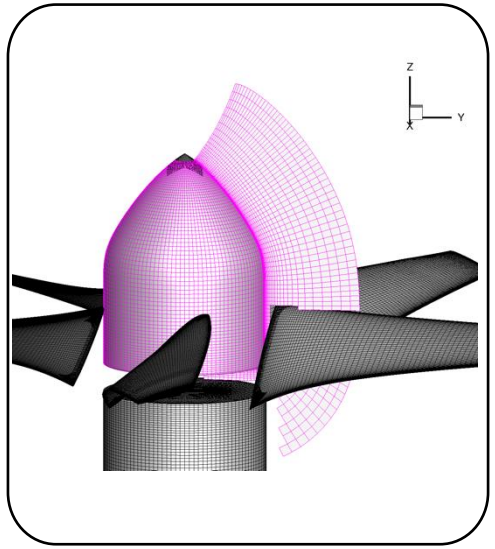
Volume grid generation

- Hyperbolic Near-Body (NB)
- Cartesian Off-Body (OB)



Domain connectivity

- Grid point blanking
- Donor stencil search

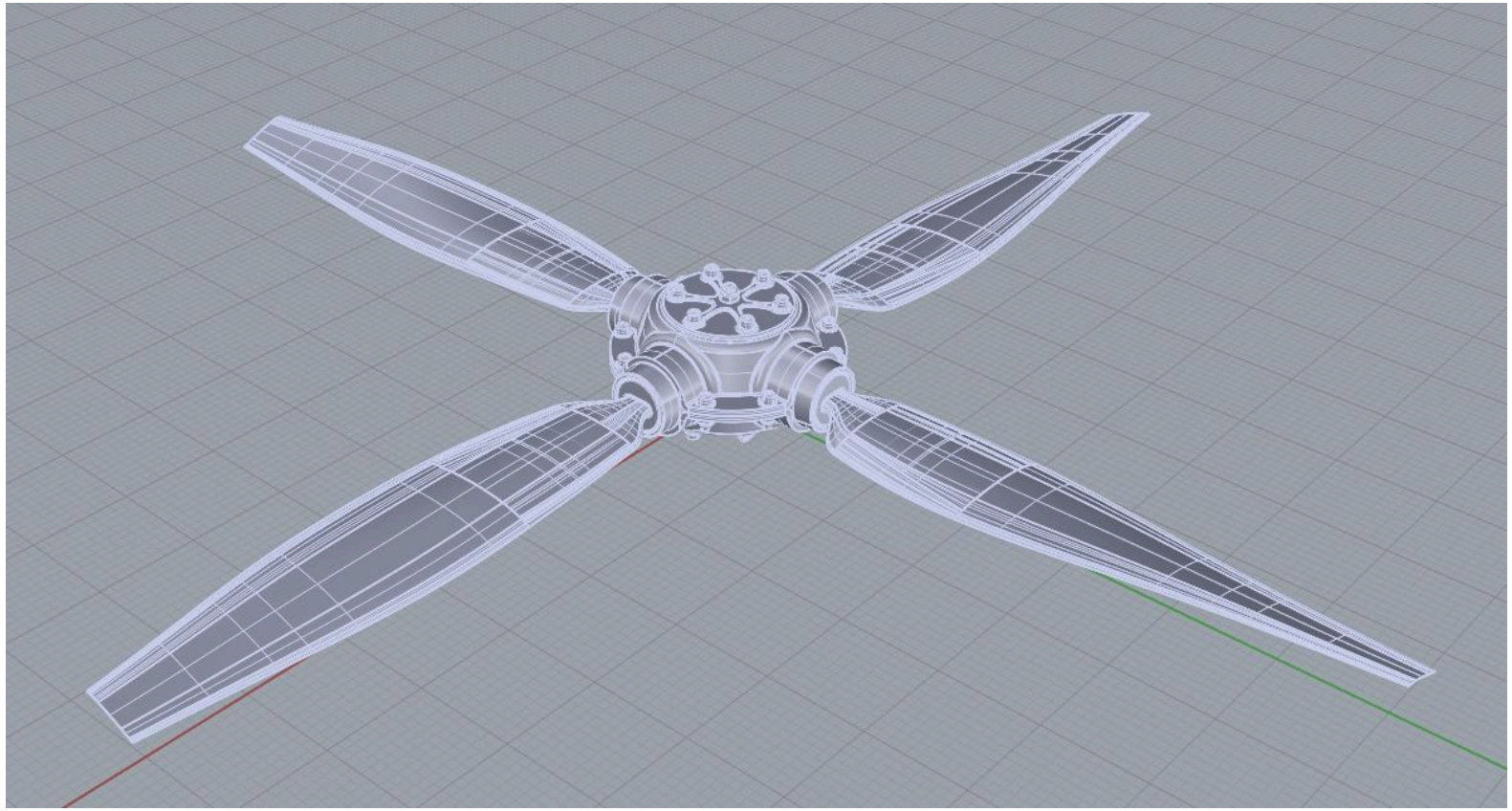


Practical example: ROAMX rotor + stand



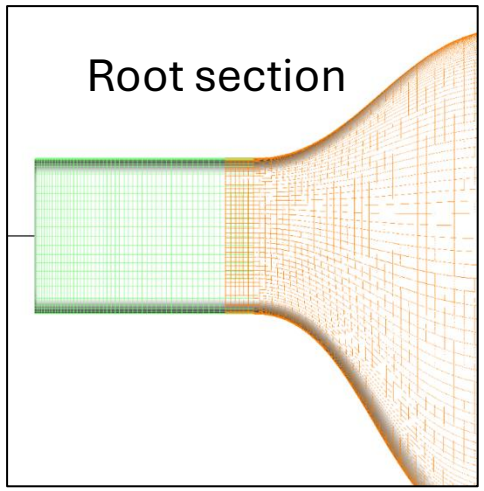
Ingenuity blades mounted in the Planetary Aeolian Laboratory (**PAL**) test stand

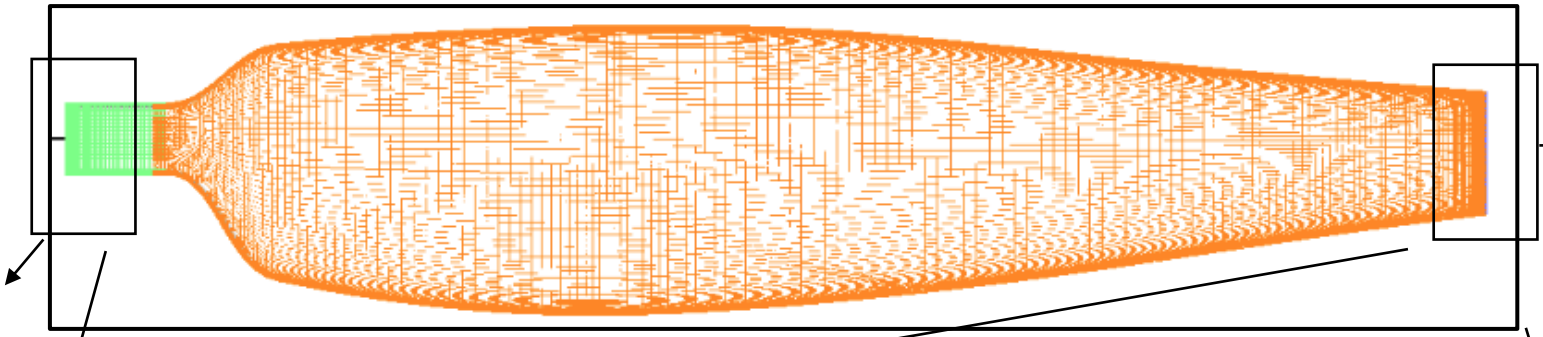
ROAMX Project:
Rotor Optimization for the Advancement of Mars eXploration

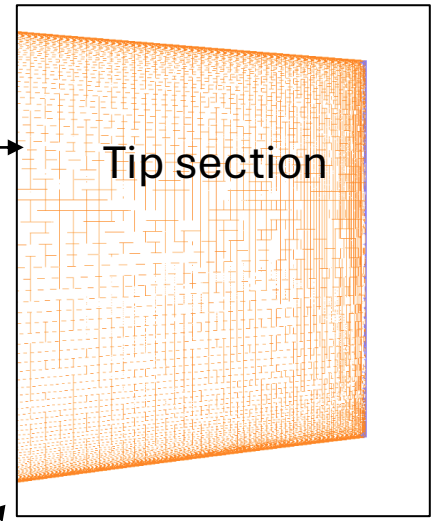


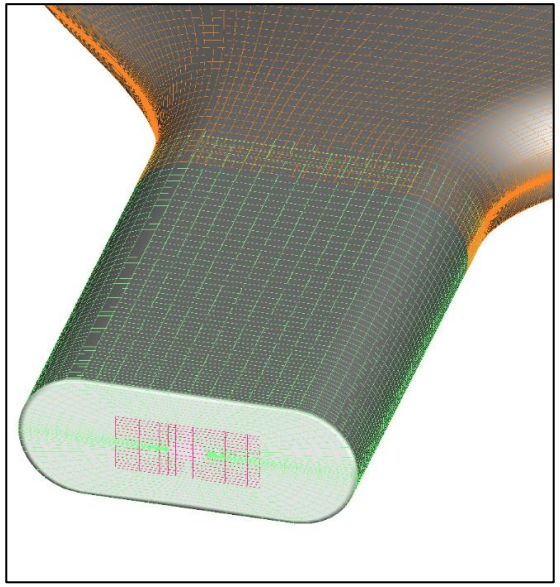
CAD of ROAMX blade and hub

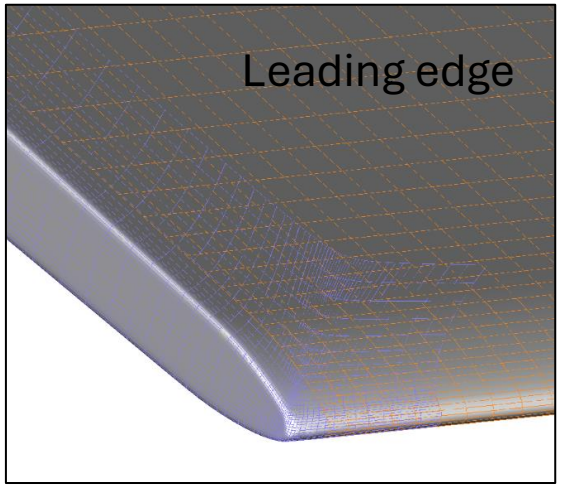
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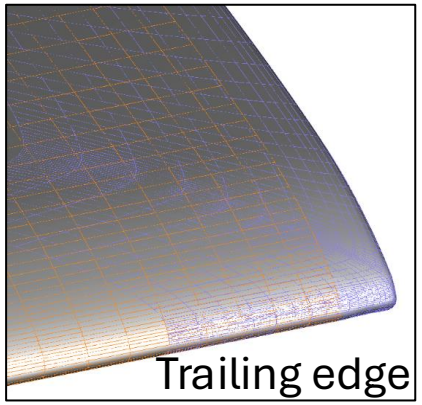


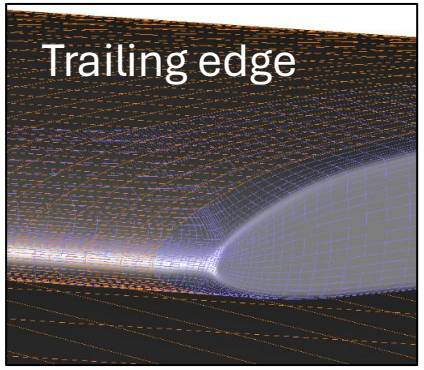


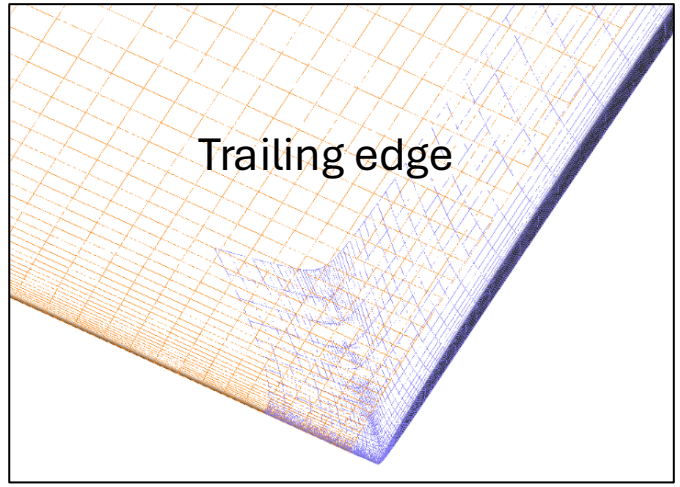






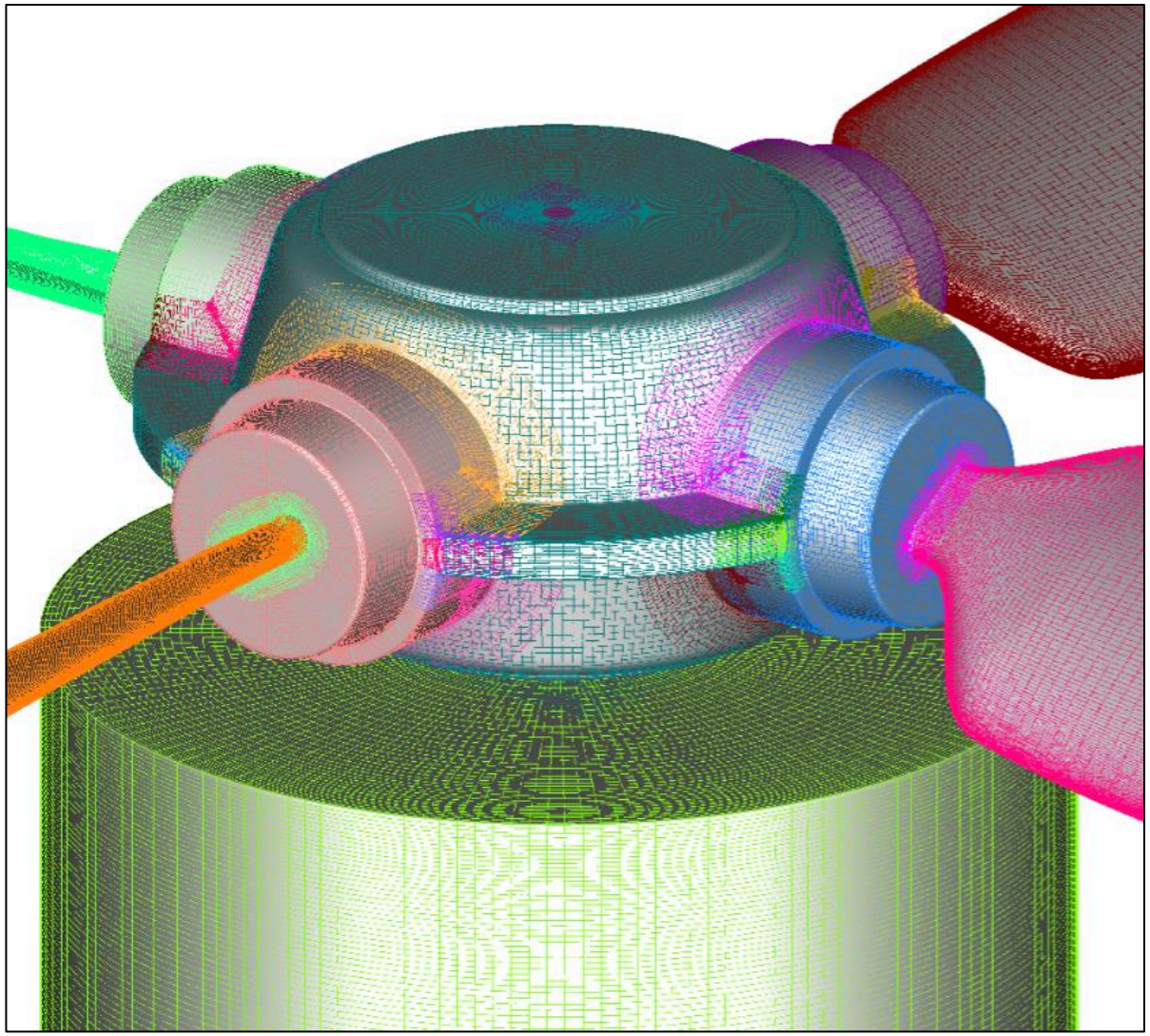


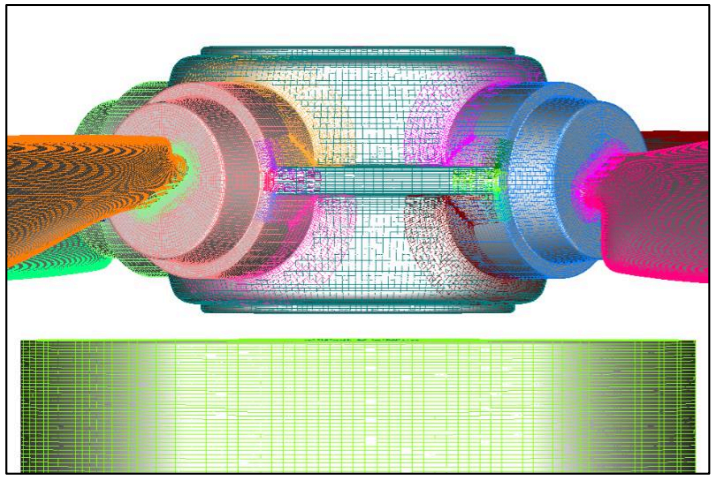


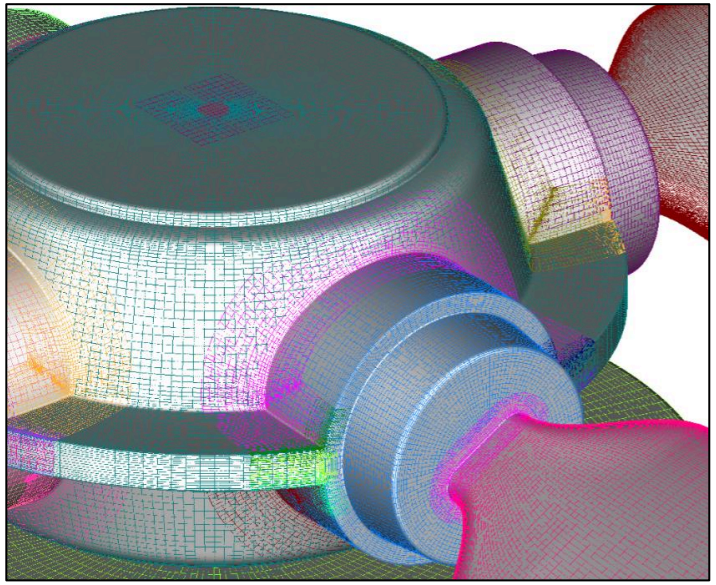


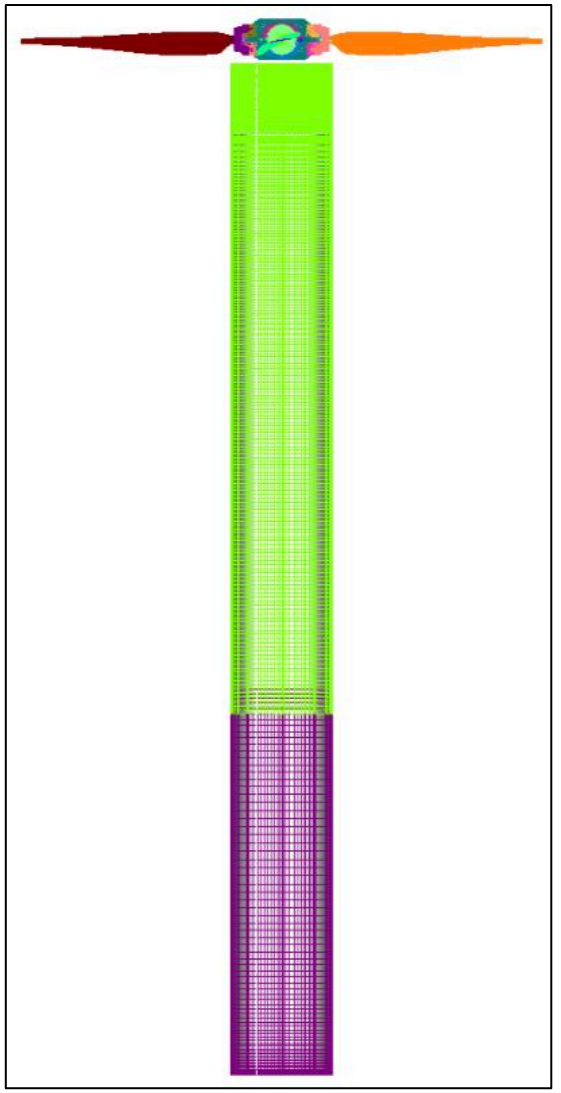
Four grids per blade: main blade, cap grip for the tip, a root section grid, and axis cap grid.
Isolated rotor simulations: 16 NB grid points with 16 NB million grid points and 114 NB+OB million grid points

Practical example: ROAMX rotor + stand



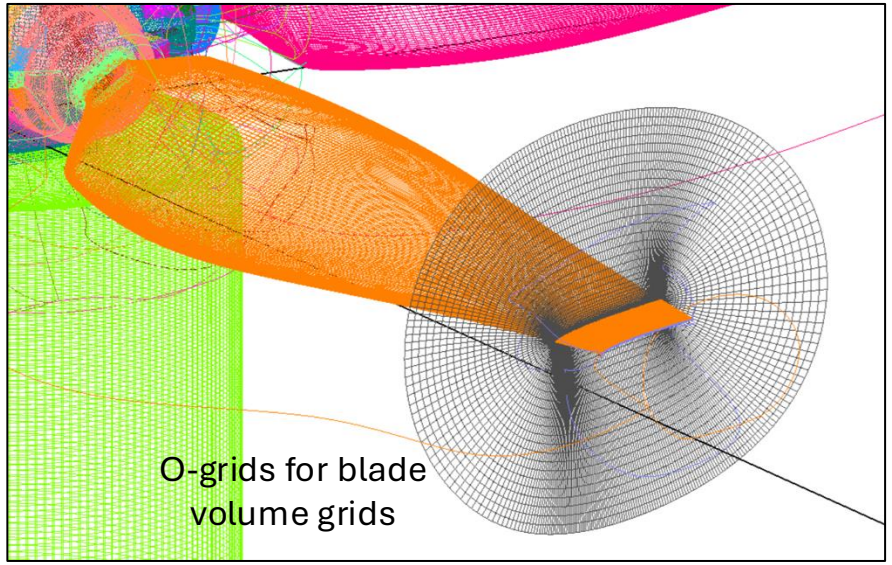


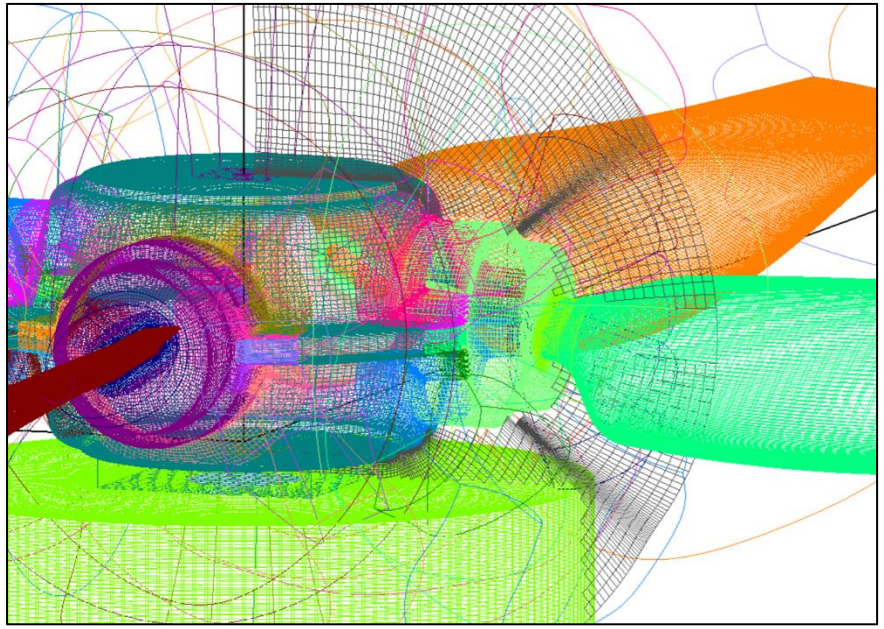


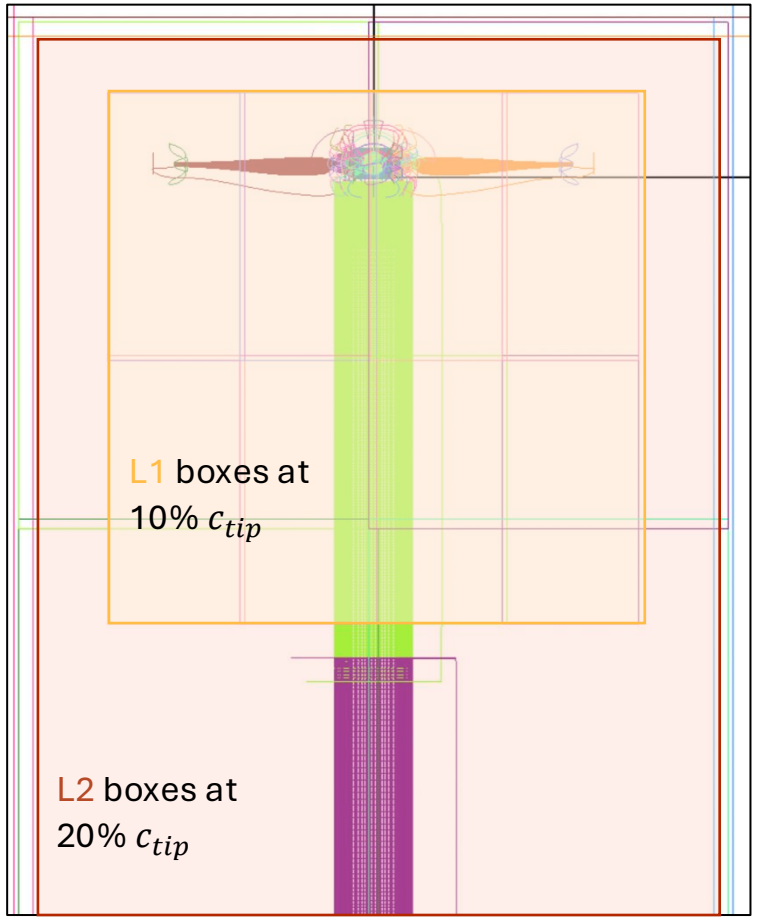


55 NB grids, 41 million NB points, 175 million total grid points

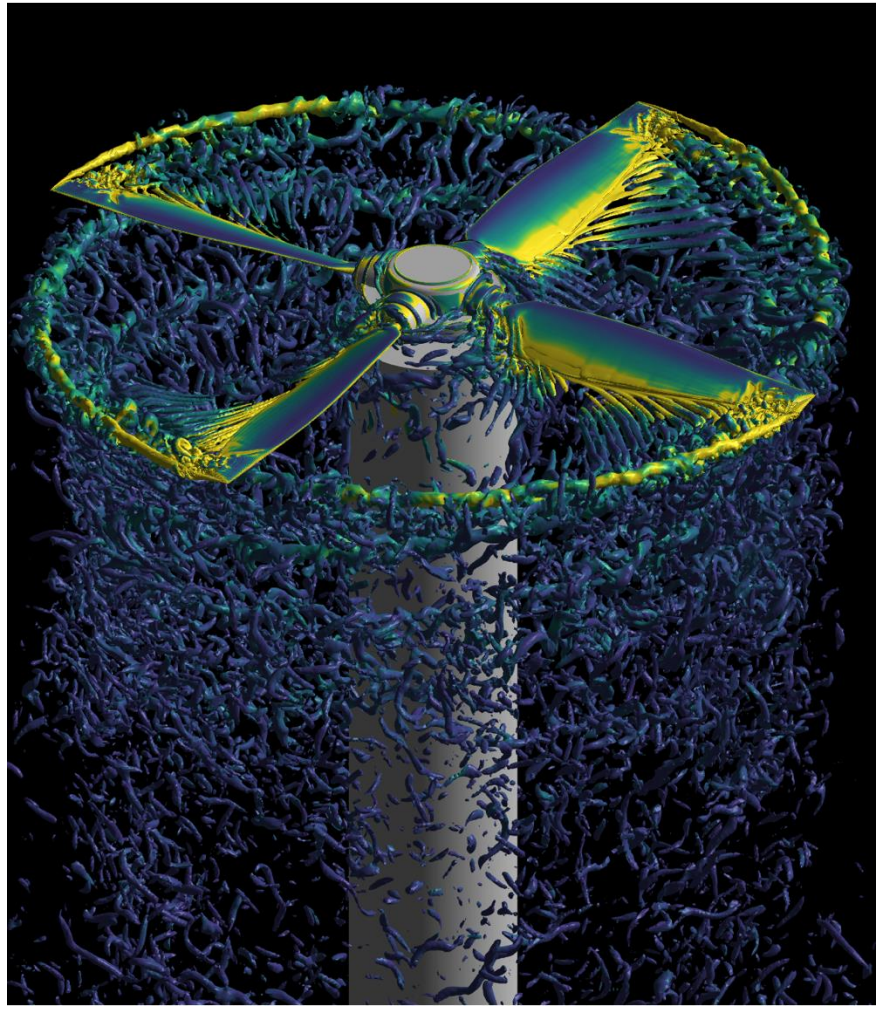
Practical example: ROAMX rotor + stand







Simulations run with 1600 processors,
~3.5h per rotor revolution.



OVERFLOW simulation of ROAMX blades. Q-criterion iso-surfaces colored with vorticity magnitude

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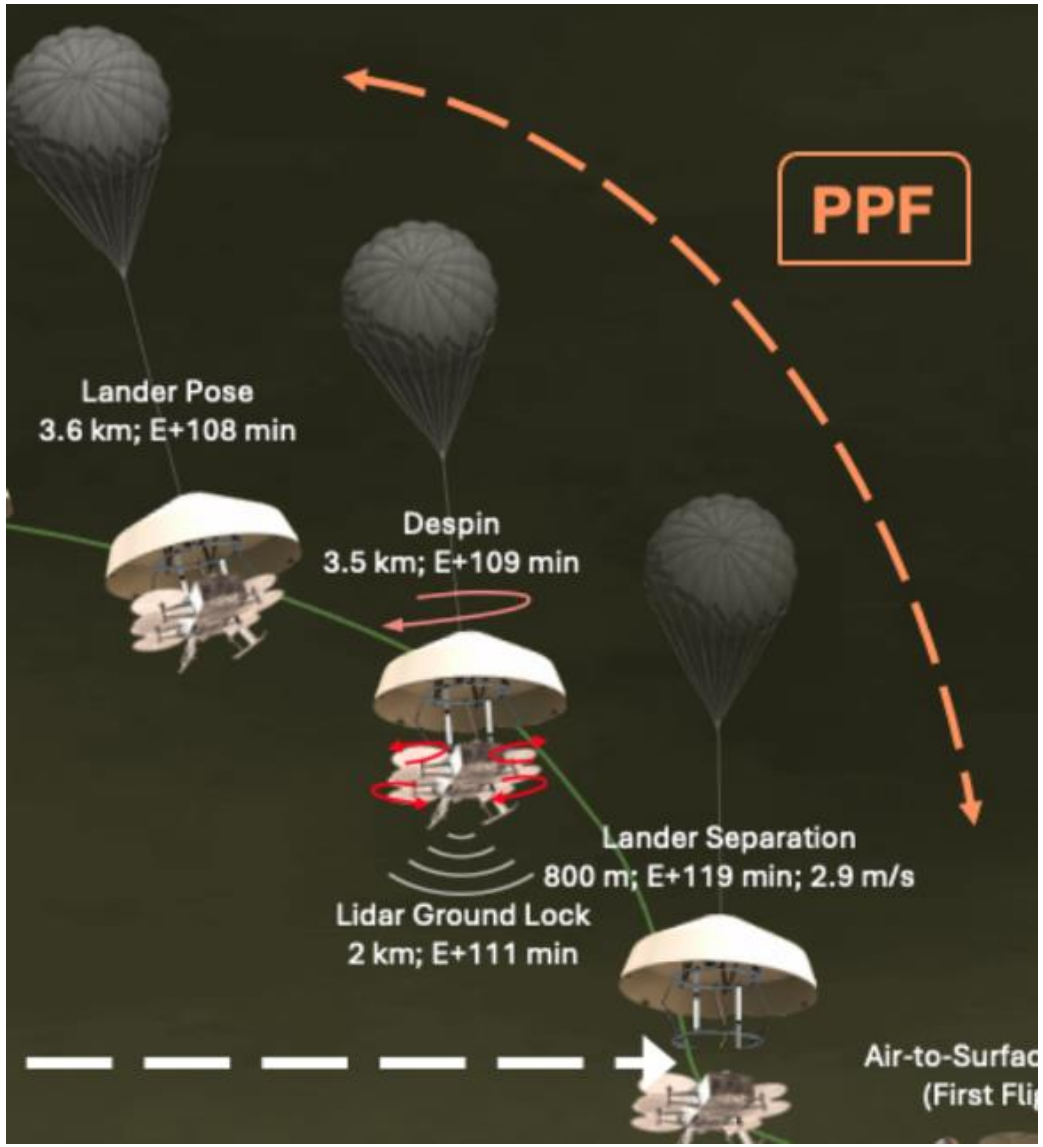
Dragonfly Rotorcraft Lander

- Preparation for Powered Flight (PPF)
- NFAC PPF: OVERFLOW validation
- Titan PPF

Skyfall Helicopters

- Mission Overview
- Skyfall: overset grids
- Skyfall: hover solution

Dragonfly: Preparation for Powered Flight (PPF)



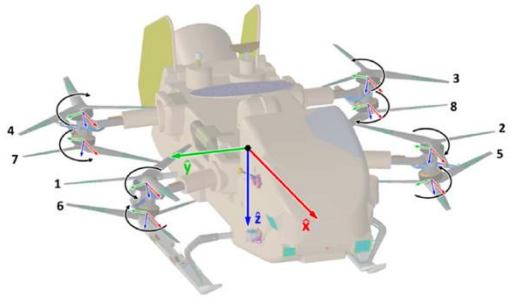
At around 1 km above the surface of Titan, the Preparation for Powered Flight (**PPF**) sequence begins.

- First, the lander is lowered out of the backshell to expose all rotors.
- The rotors perform the so-called **de-spin maneuver** to remove any residual spin and it is maintained until release from the backshell.

The lander is released once a safe state in which Transition to Powered Flight (**TPF**) is possible.

- The TPF maneuver begins when the lander is released from the backshell and finishes when a steady-state descent condition is attained.

During powered descent the lander will search for safe landing sites using onboard sensors.





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NFAC PPF: OVERFLOW validation

Testing in the National Full-Scale Aerodynamics Complex (NFAC) 80x120 ft wind tunnel to simulate PPF conditions was completed in February 2024.

- NFAC's 80x120 ft was selected for Dragonfly PPF testing for its availability and to reduce wind tunnel wall effects on data quality.
- The objective of the test was to measure the combined aerodynamic/propulsive forces and moments from the lander and backshell at scaled PPF conditions.

Aerial view of NFAC wind tunnel, NASA ARC.

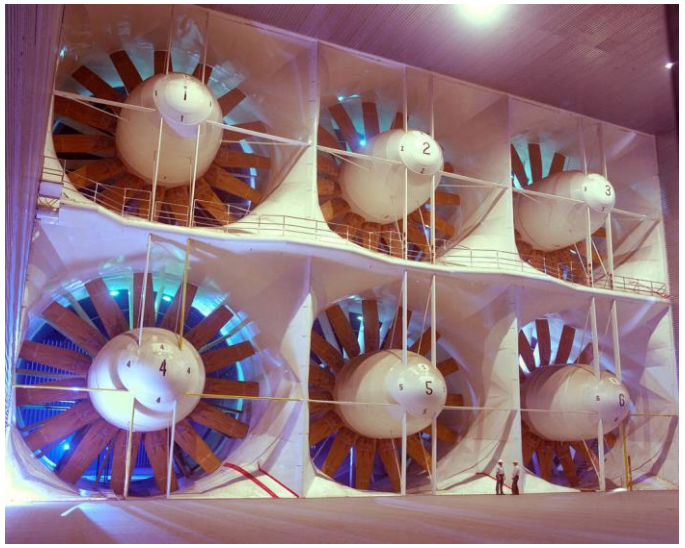


Dragonfly lander+backshell installed in NFAC.





NFAC 80x120ft wind tunnel drive fans.



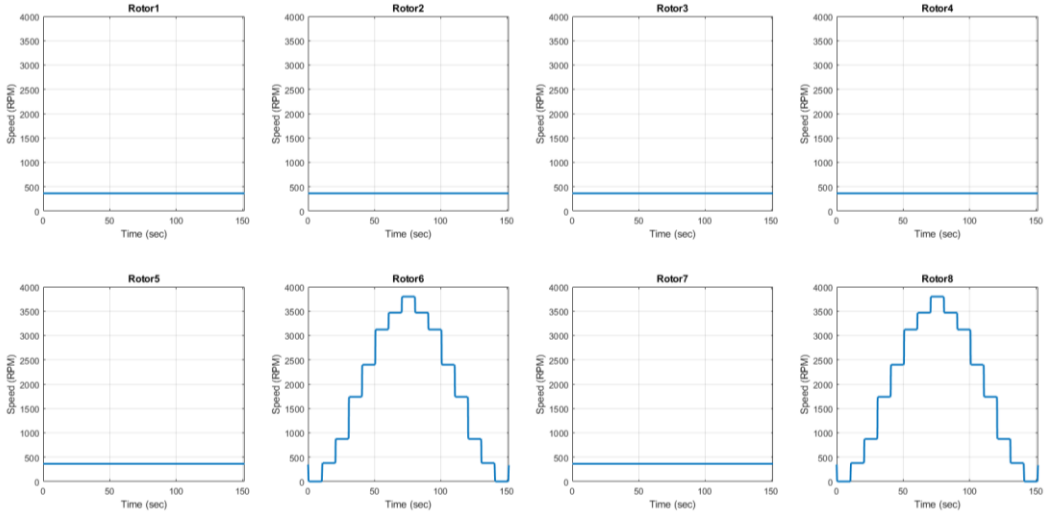
NFAC PPF: selected cases for OVERFLOW

RN-PN-SN	Test Section Velocity (m/s)	Lander Pitch (deg)	Lander Roll (deg)	Rotor Sequence	RPM_ref1	RPM_ref2
19-7-6	5.2354	90	0	Q7a	3470	365
17-7-6	6.8871	90	0	Q7a	3470	365
18-35-6	6.8733	130	10	Q7a	3470	365
18-43-8	6.9164	130	30	Q7a	3470	365
19-7-3	5.2820	90	0	Q7a	1740	365
19-7-4	5.2513	90	0	Q7a	2402	365
19-7-5	5.2301	90	0	Q7a	3124	365

OVERFLOW (blade-resolved) will be compared with NFAC test data and with mid-fidelity STAR-CCM+ simulations (actuator disk).

- Two test section velocities at max RPM and lander pitch and roll (P/R) at 90/0.
- Effect of lander pitch and roll at max RPM and same test section velocity $V_\infty = 6.9$ m/s.
- RPM sweep with P/R 90/0 and same test section velocity $V_\infty = 5.2$ m/s.

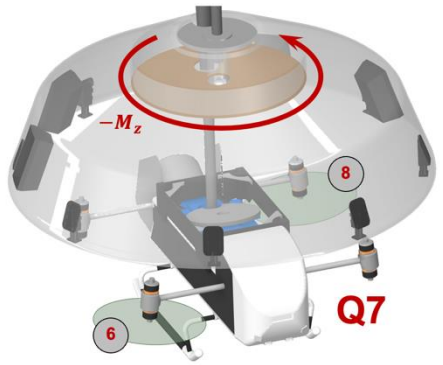
Q7a Rotor Sequence: rotor 6 (**R6**) and rotor 8 (**R8**) for de-spin



RPM set points:

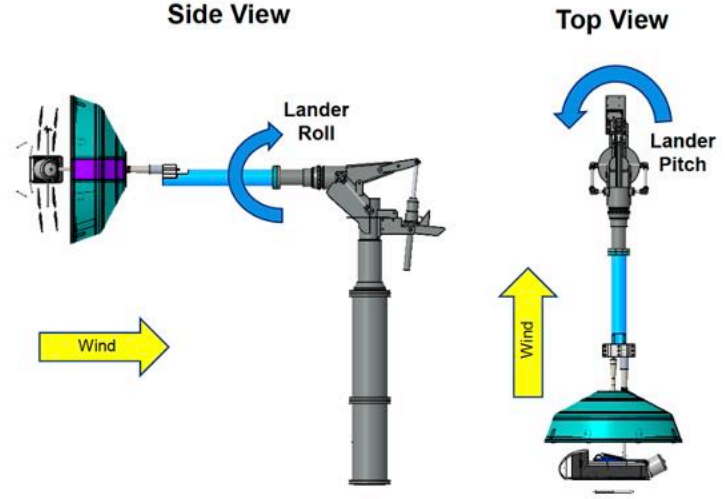
Q7

380
875
1740
2400
3124
3470
3800

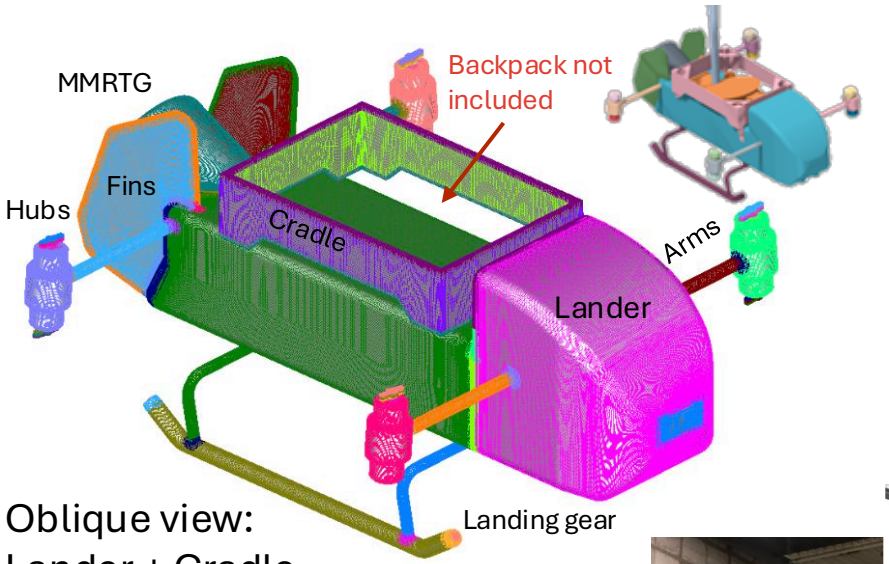


Diagonally opposed lower rotors used to create yaw moment. Desired $-M_z$ in lander body axis for Q7a.

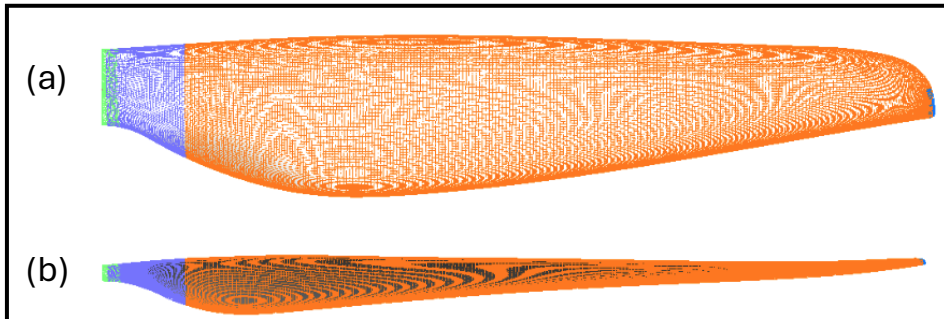
Lander pitch and roll in NFAC



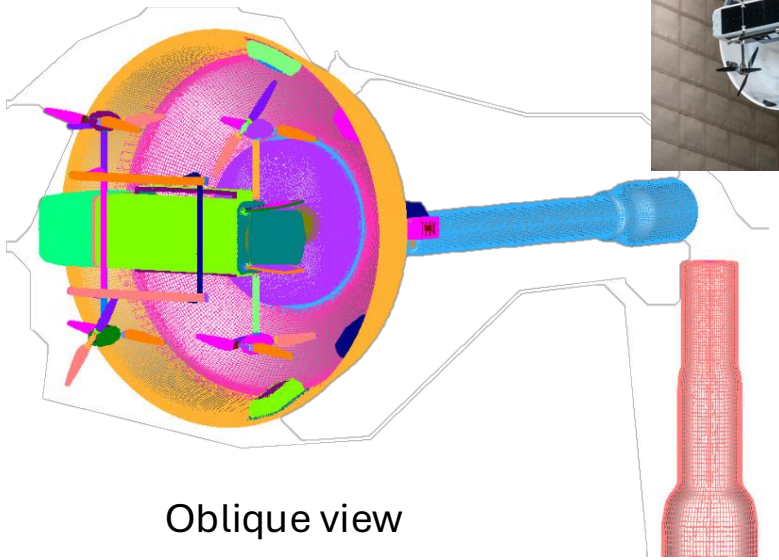
NFAC PPF: near-body grids



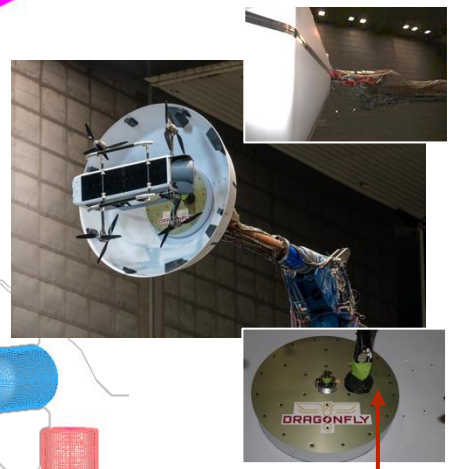
Oblique view:
Lander + Cradle



KDE blade near-body grids; (a) top view, (b) side view.

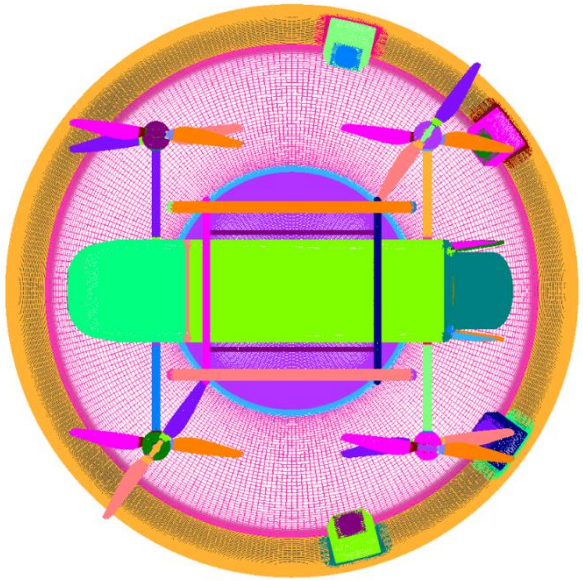


Oblique view

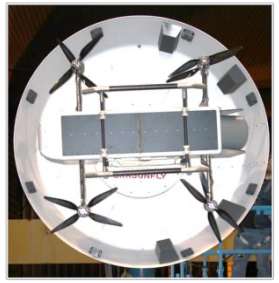


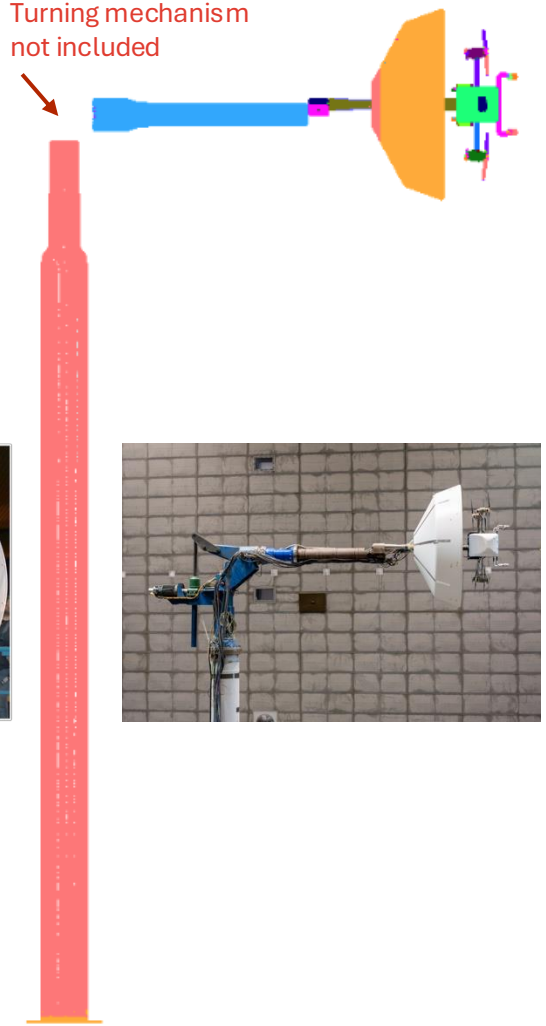
Small gap
Backshell mesh is sealed

$D = 2.581$ m



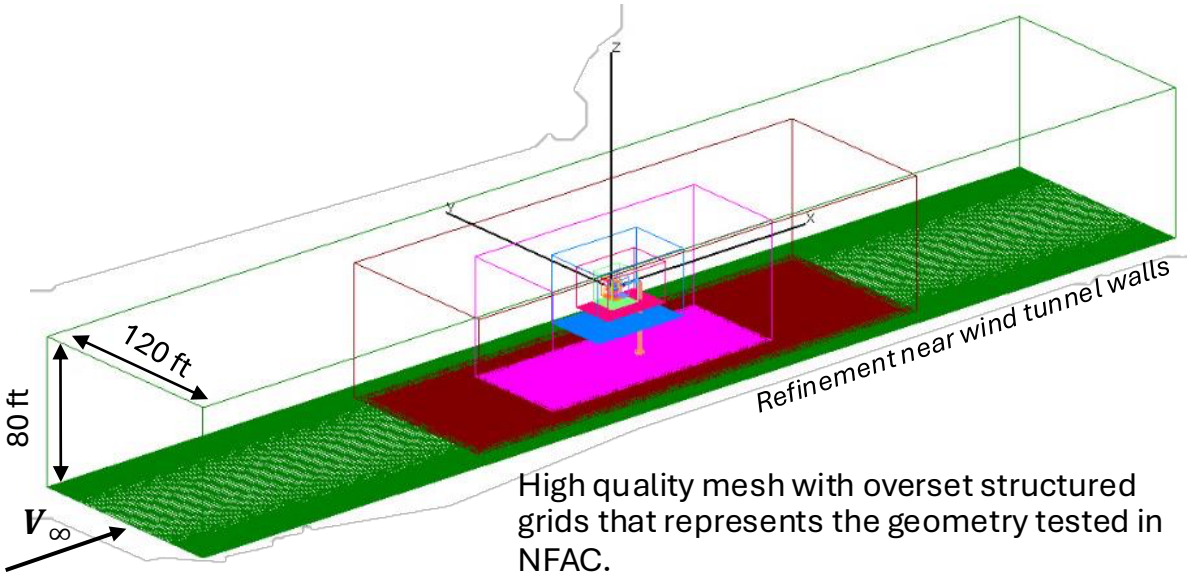
Front view

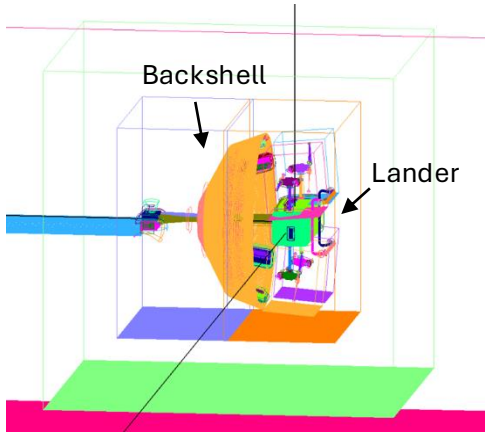




Side view

NFAC PPF: wind tunnel grids





View of refinement regions near rotors and backshell

High quality mesh with overset structured grids that represents the geometry tested in NFAC.

Geometry includes extended NFAC test section.

Refinement regions to accurately solve rotors and backshell wakes.

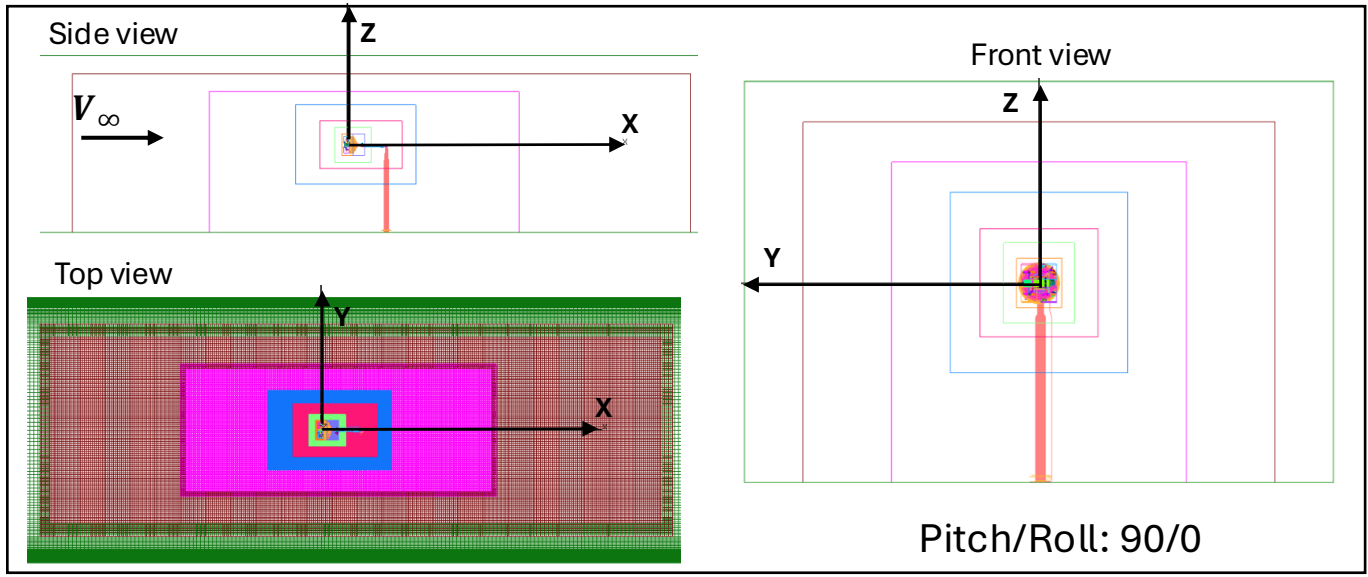
- Refinement regions move with the model as pitch and roll are applied.
- Domain connectivity performed accordingly.

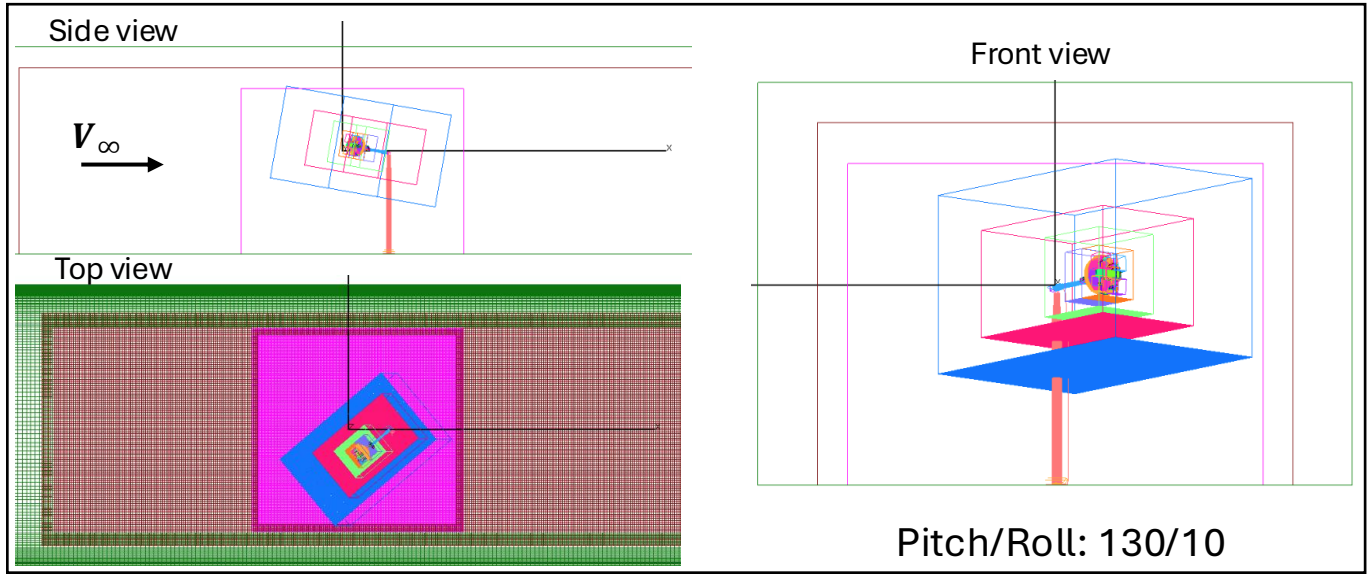
228 grids for the model + stand, 11 grids for refinement regions and wind tunnel.

Total mesh size: 280 million grid points.

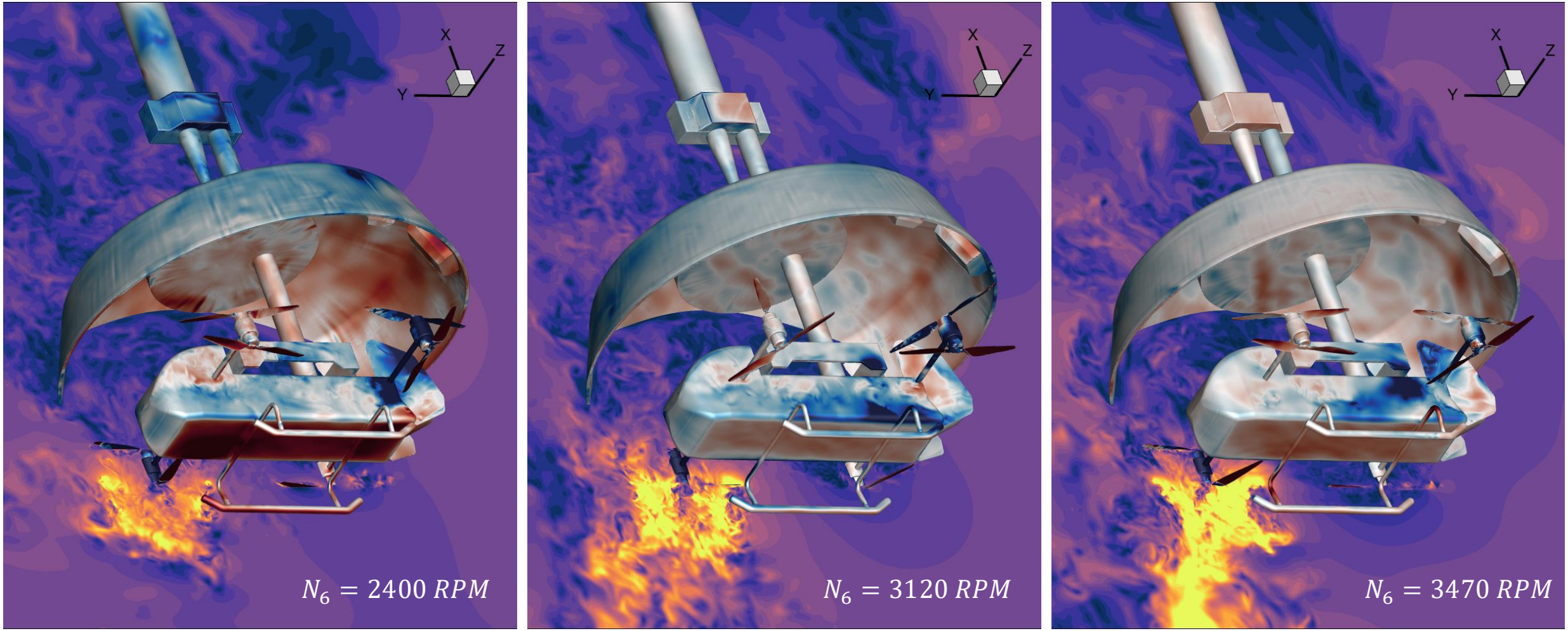
Simulations ran with 2520 processors.

- 1 rotor revolution @ MAX rpm ~ 5h.





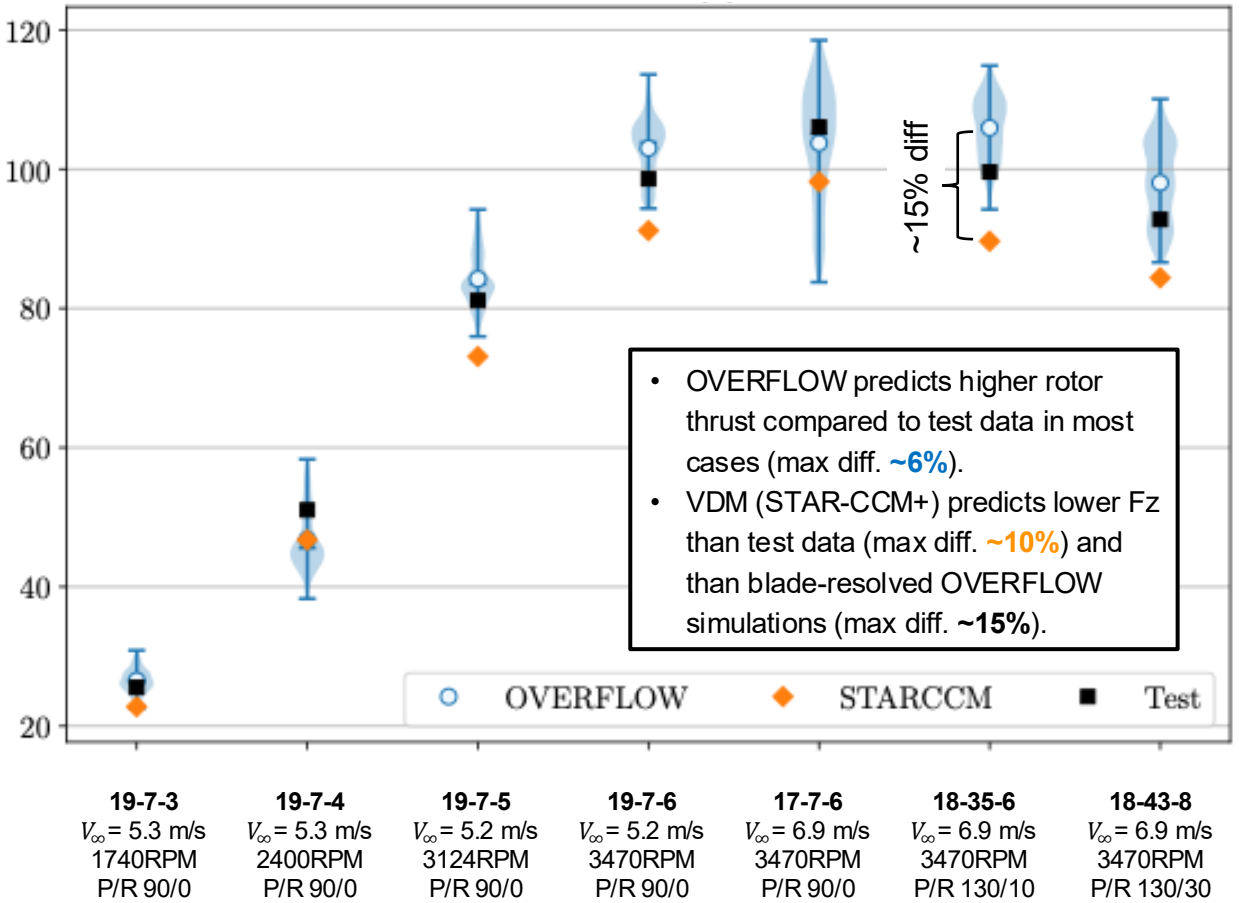
NFAC PPF: flow visualization



RPM sweep test cases 19-7-4, 19-7-5, and 19-7-6, conducted at a tunnel velocity of $V_\infty \approx 5.2\ m/s$ under axial descent conditions with P/R = 90°/0°. Each image shows the instantaneous pressure coefficient on the Dragonfly NFAC test model, along with a velocity magnitude slice through the right-side rotor plane, highlighting the wake from rotor 6.

NFAC PPF: rotor 8 performance

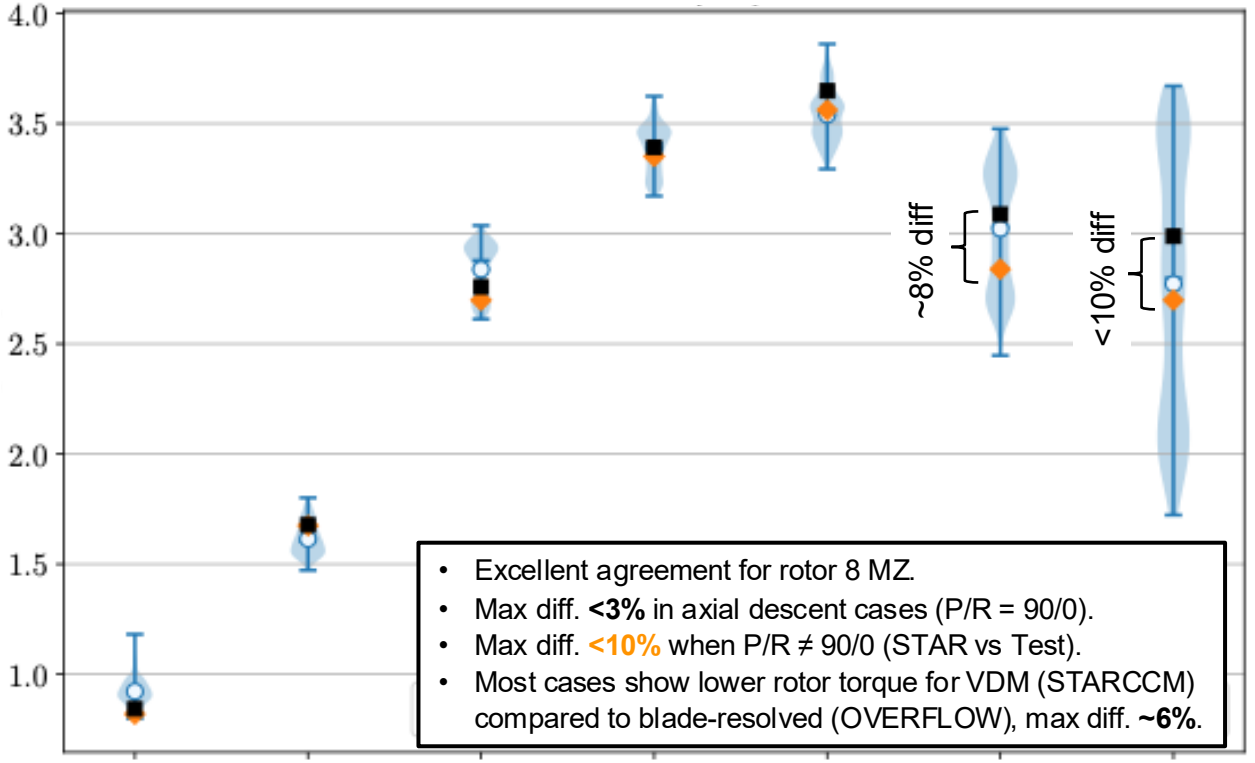
Rotor 8 Fz [N]



- OVERFLOW predicts higher rotor thrust compared to test data in most cases (max diff. **~6%**).
- VDM (STAR-CCM+) predicts lower Fz than test data (max diff. **~10%**) and than blade-resolved OVERFLOW simulations (max diff. **~15%**).

19-7-3	19-7-4	19-7-5	19-7-6	17-7-6	18-35-6	18-43-8
V_∞ = 5.3 m/s 1740RPM P/R 90/0	V_∞ = 5.3 m/s 2400RPM P/R 90/0	V_∞ = 5.2 m/s 3124RPM P/R 90/0	V_∞ = 5.2 m/s 3470RPM P/R 90/0	V_∞ = 6.9 m/s 3470RPM P/R 90/0	V_∞ = 6.9 m/s 3470RPM P/R 130/10	V_∞ = 6.9 m/s 3470RPM P/R 130/30

Rotor 8 Mz [Nm]



- Excellent agreement for rotor 8 MZ.
- Max diff. **<3%** in axial descent cases (P/R = 90/0).
- Max diff. **<10%** when P/R ≠ 90/0 (STAR vs Test).
- Most cases show lower rotor torque for VDM (STARCCM) compared to blade-resolved (OVERFLOW), max diff. **~6%**.

19-7-3	19-7-4	19-7-5	19-7-6	17-7-6	18-35-6	18-43-8
V_∞ = 5.3 m/s 1740RPM P/R 90/0	V_∞ = 5.3 m/s 2400RPM P/R 90/0	V_∞ = 5.2 m/s 3124RPM P/R 90/0	V_∞ = 5.2 m/s 3470RPM P/R 90/0	V_∞ = 6.9 m/s 3470RPM P/R 90/0	V_∞ = 6.9 m/s 3470RPM P/R 130/10	V_∞ = 6.9 m/s 3470RPM P/R 130/30

OUTLINE

Numerical Approach

- High-Fidelity CFD solver: NASA's OVERFLOW
- Overset Grids: NASA's Chimera Grid Tools
- Practical example: ROAMX rotor + stand

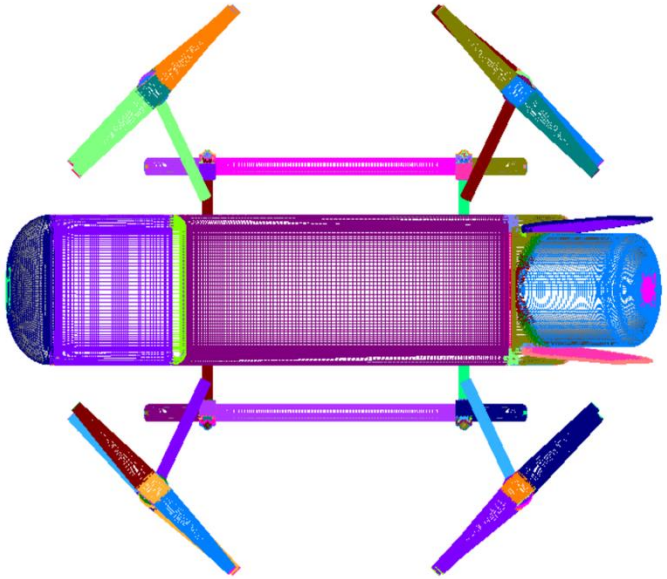
Dragonfly Rotorcraft Lander

- Preparation for Powered Flight (PPF)
- NFAC PPF: OVERFLOW validation
- Titan PPF

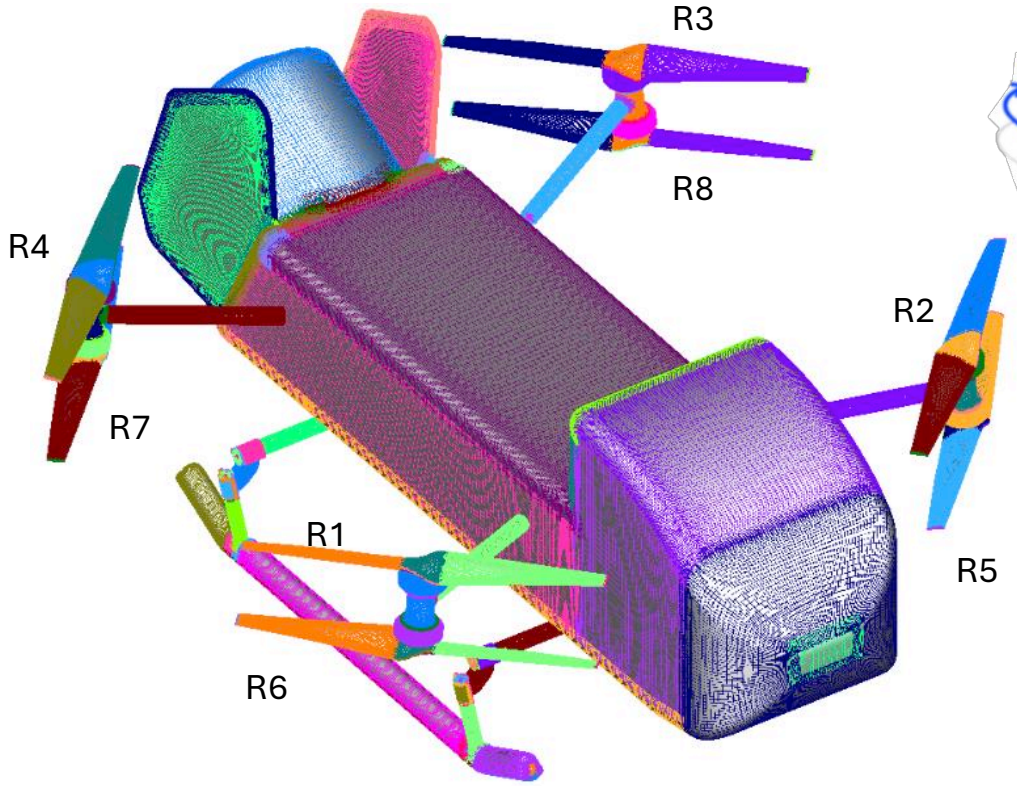
Skyfall Helicopters

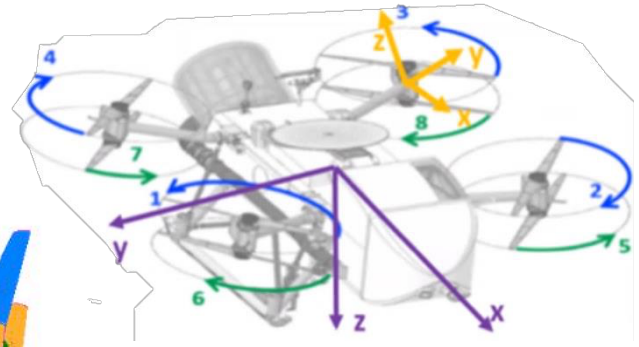
- Mission Overview
- Skyfall: overset grids
- Skyfall: hover solution

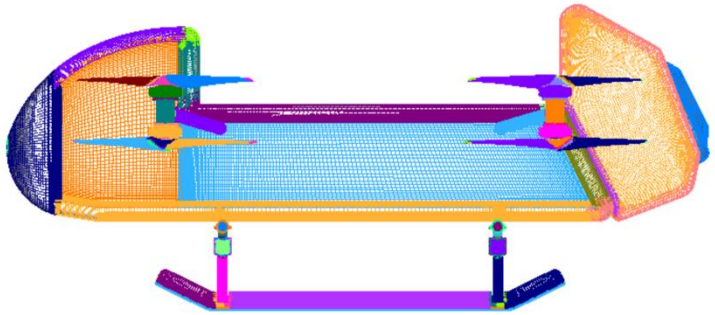
Titan PPF: lander mesh



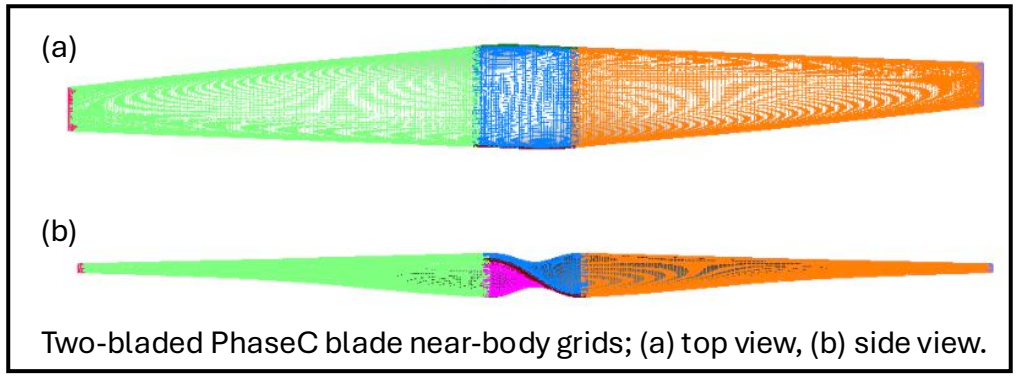
Top view Dragonfly lander DC0



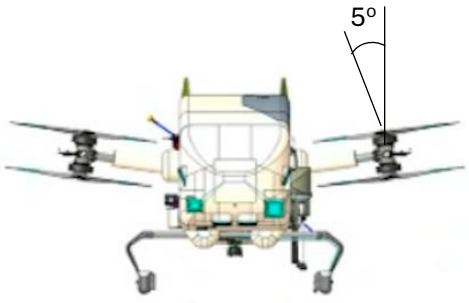




Side view Dragonfly lander DC0

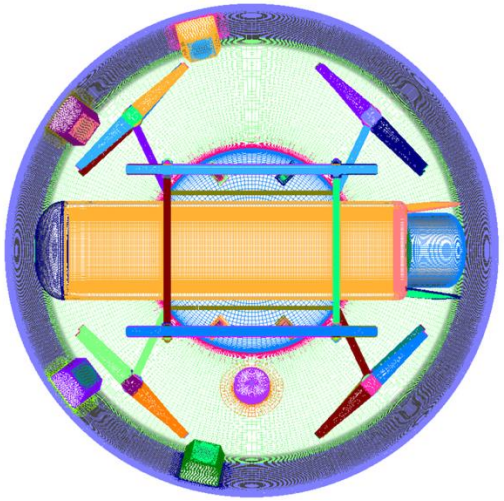


Two-bladed PhaseC blade near-body grids; (a) top view, (b) side view.

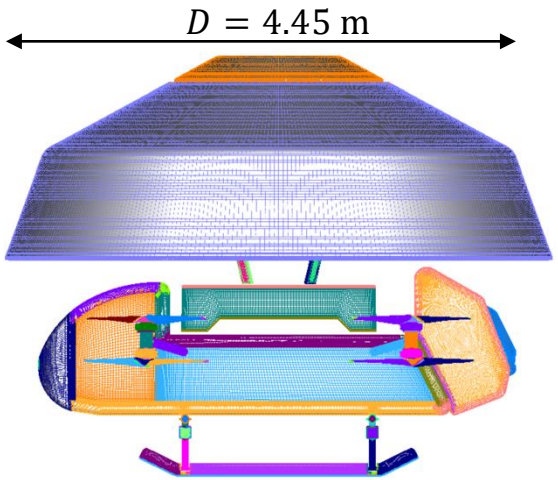


DC0 geometry has 5deg rotor cant around X axis (same for upper and lower rotors).

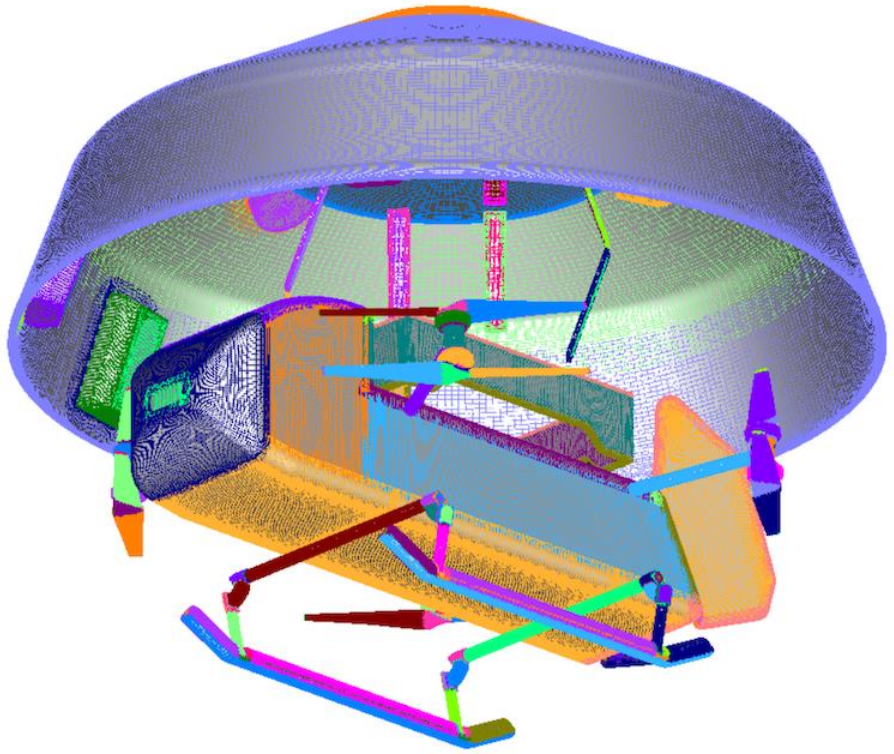
Titan PPF: lander + backshell mesh



Bottom view PPF with lander DC0



Side view PPF with lander DC0



Oblique view PPF with lander DC0

Simplified geometry for OVERFLOW includes:

- Lander geometry
- Backshell
- Backshell additional components
- Cradle
- Connecting rods backshell - lander

Simplified geometry does not include:

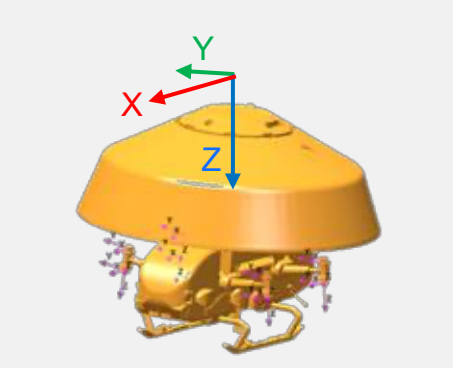
- HGA, LGA, MGA, probes, arm fins

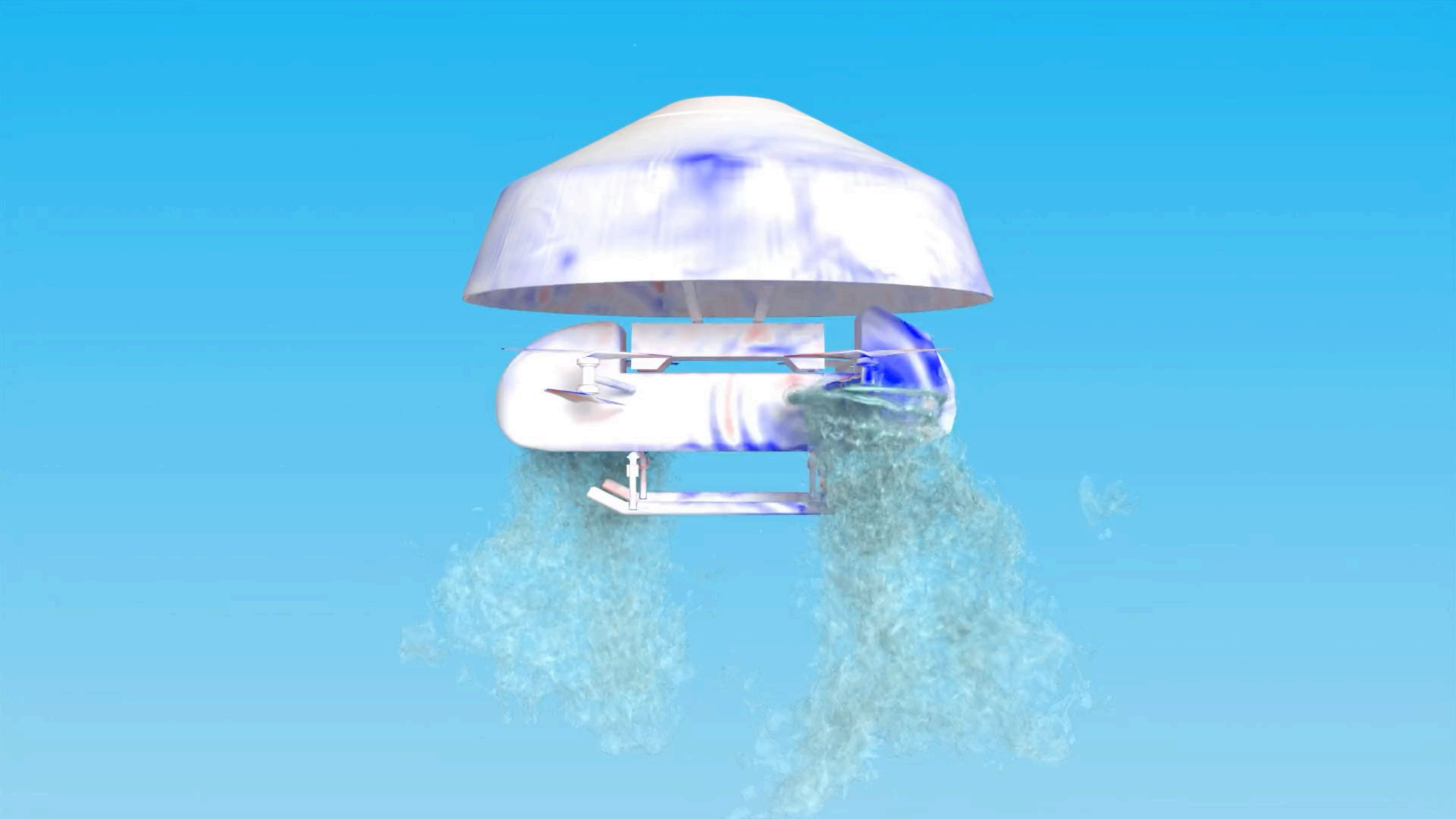
342 NB grids with **137** million NB grid points.

Total mesh size: **400 to 450 million grid points**.

Simulations ran with 2520 processors.

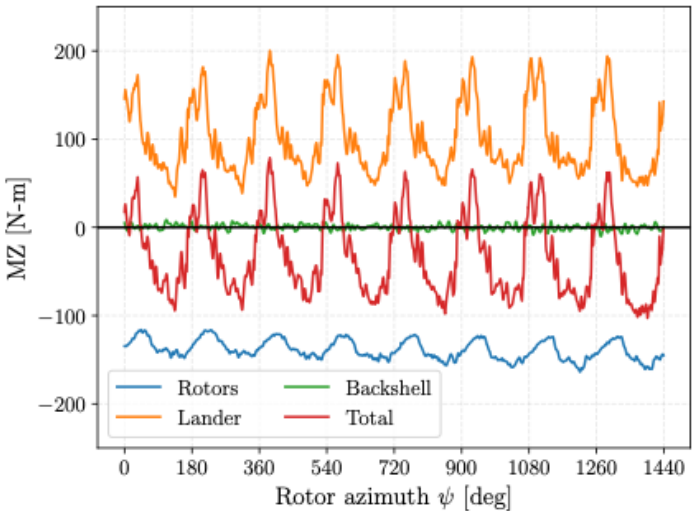
- 1 rotor revolution @ MAX rpm ~ 8.5h.





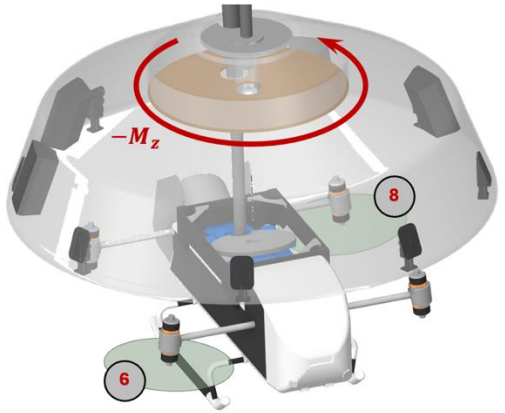
Titan PPF: unsteady loads

Pure descent 1

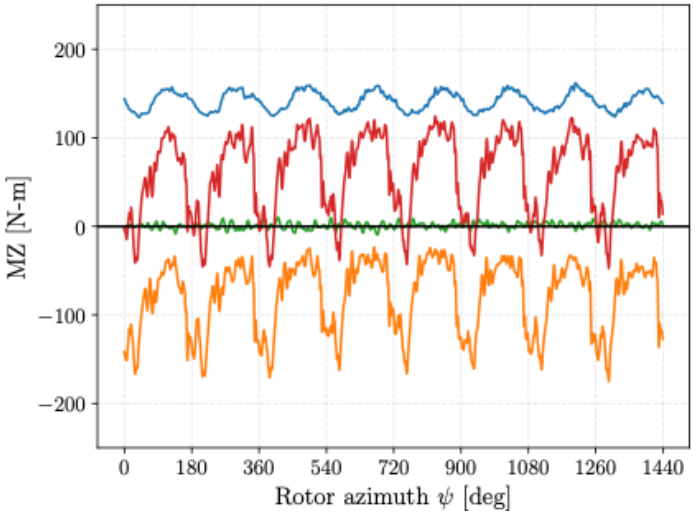


Rotors **6** and **8** for de-spin

- Expected $M_z < 0$

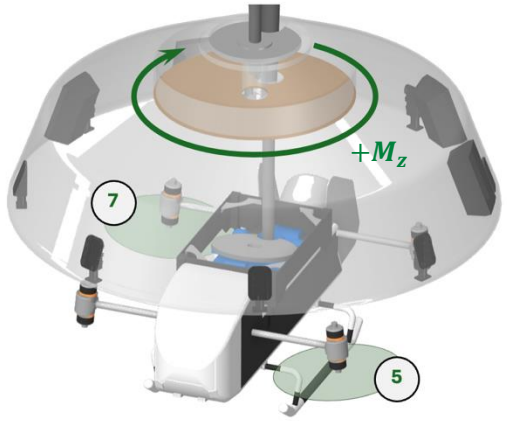


Pure descent 2



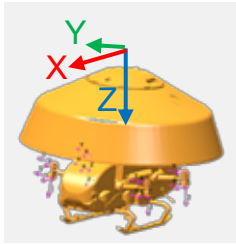
Rotors **5** and **7** for de-spin

- Expected $M_z > 0$



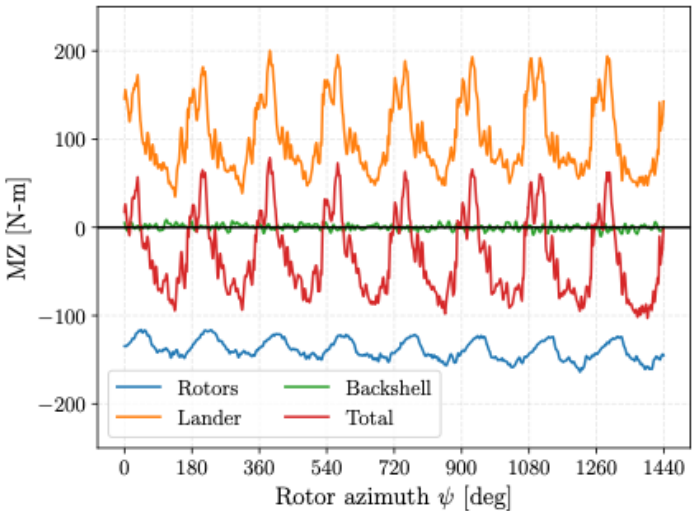
Selected PPF cases for OVERFLOW:

- **Pure descent 1**
  - All rotors 100 RPM except **R6&R8** at 1100 RPM
  - V_∞ = (0, 0, 3) m/s
- **Design 80**
  - All rotors 100 RPM except **R6&R8** at 1100 RPM
  - V_∞ = (-0.3, 0, 2.5) m/s
- **Design 103**
  - All rotors 100 RPM except **R6&R8** at 1100 RPM
  - V_∞ = (2.9, 2.6, 2) m/s
- **Pure descent 2**
  - All rotors 100 RPM except **R5&R7** at 1100 RPM
  - V_∞ = (0, 0, 3) m/s
- **Design 164**
  - All rotors 100 RPM except **R5&R7** at 1100 RPM
  - V_∞ = (0.3, 0.3, 2.6) m/s
- **Design 192**
  - All rotors 100 RPM except **R5&R7** at 1100 RPM
  - V_∞ = (3, -2.9, 3) m/s

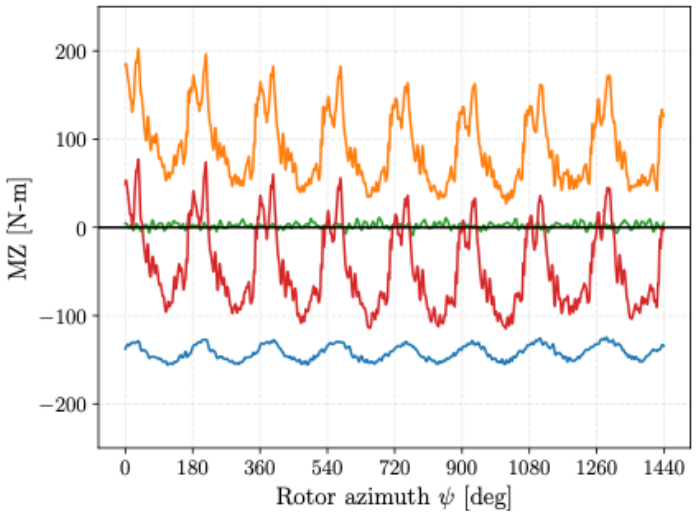


Titan PPF: unsteady loads

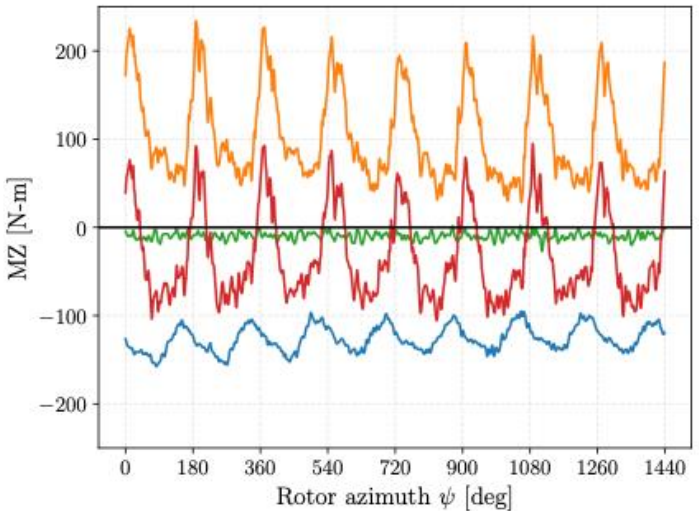
Pure descent 1



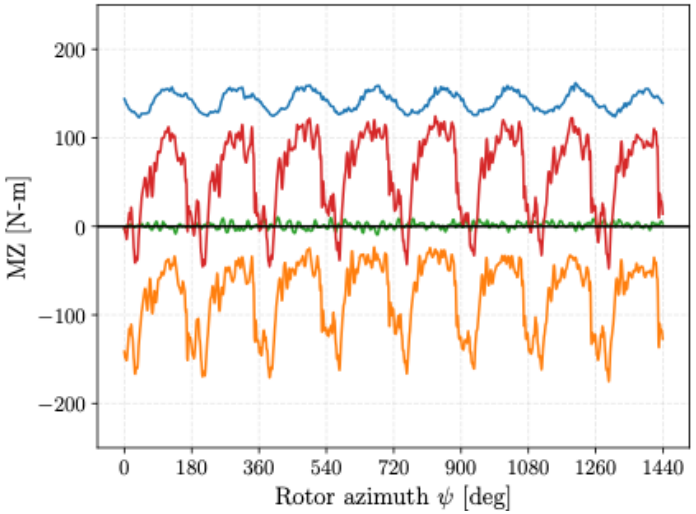
Design 80



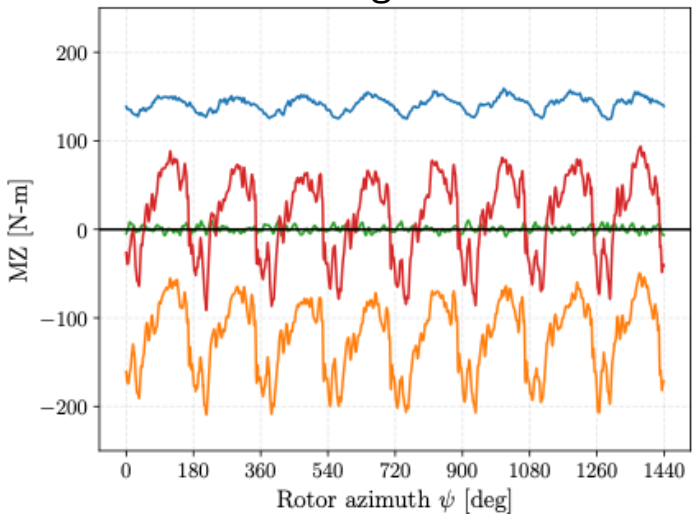
Design 103



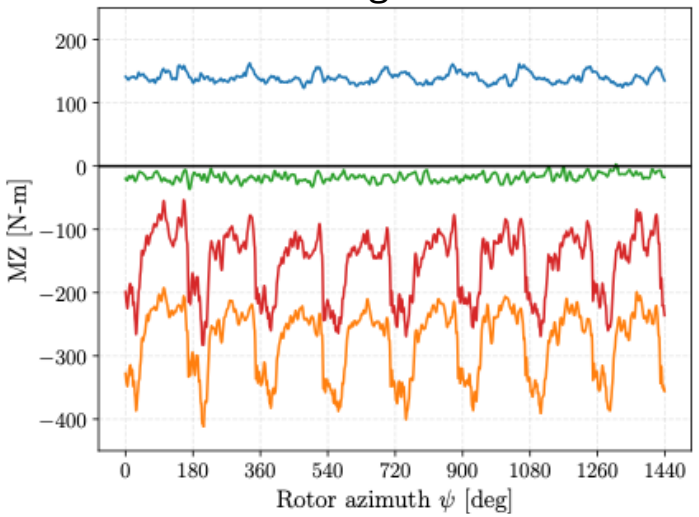
Pure descent 2



Design 164

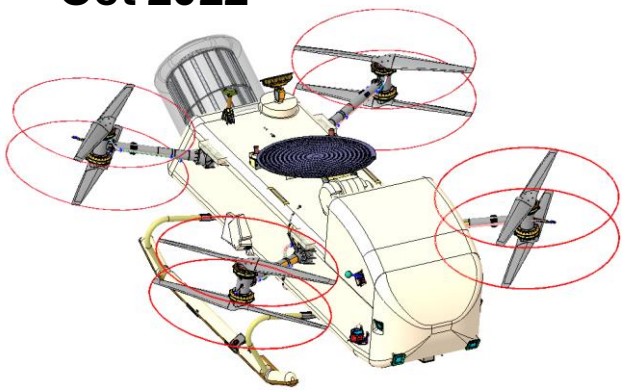


Design 192

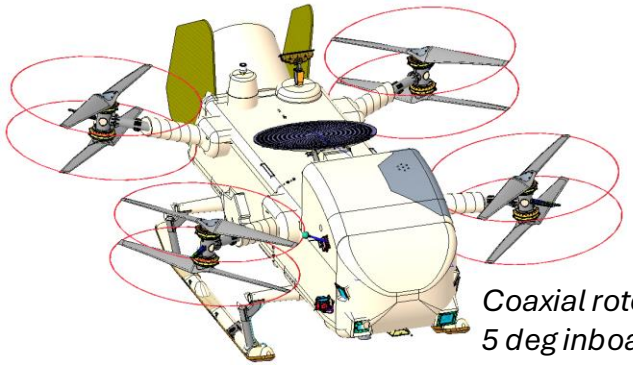


Lander configuration evolution

Oct 2022

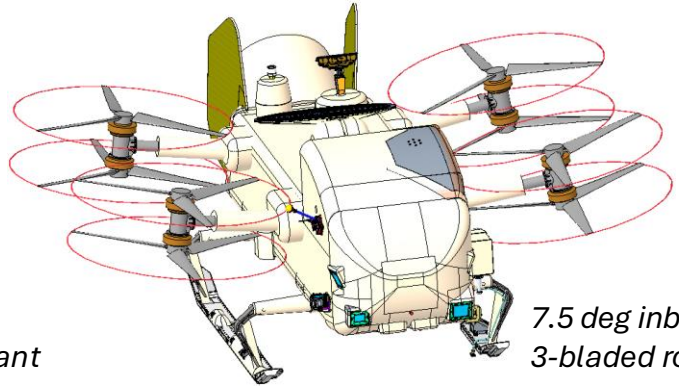


Apr 2023



Coaxial rotor pairs,
5 deg inboard rotor cant

Dec 2023



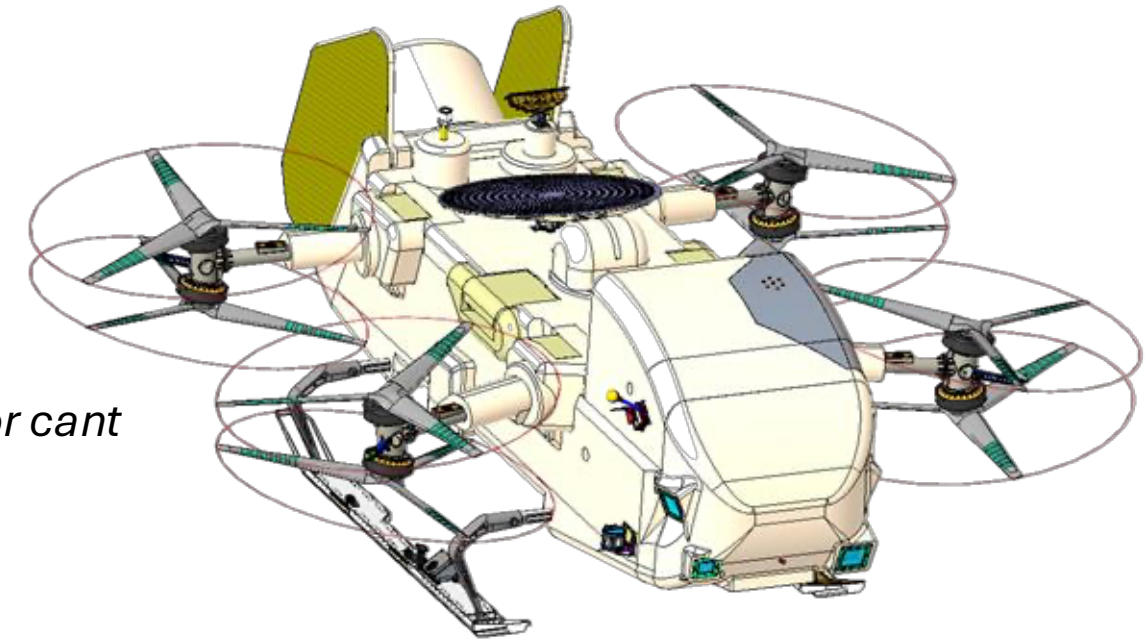
7.5 deg inboard rotor cant,
3-bladed rotors



Aug 2024 (current)

outboard upper rotor cant

Non-coaxial rotor pairs,
7.5 deg inboard rotor cant



OUTLINE

Numerical Approach

- High-Fidelity CFD solver: NASA's OVERFLOW
- Overset Grids: NASA's Chimera Grid Tools
- Practical example: ROAMX rotor + stand

Dragonfly Rotorcraft Lander

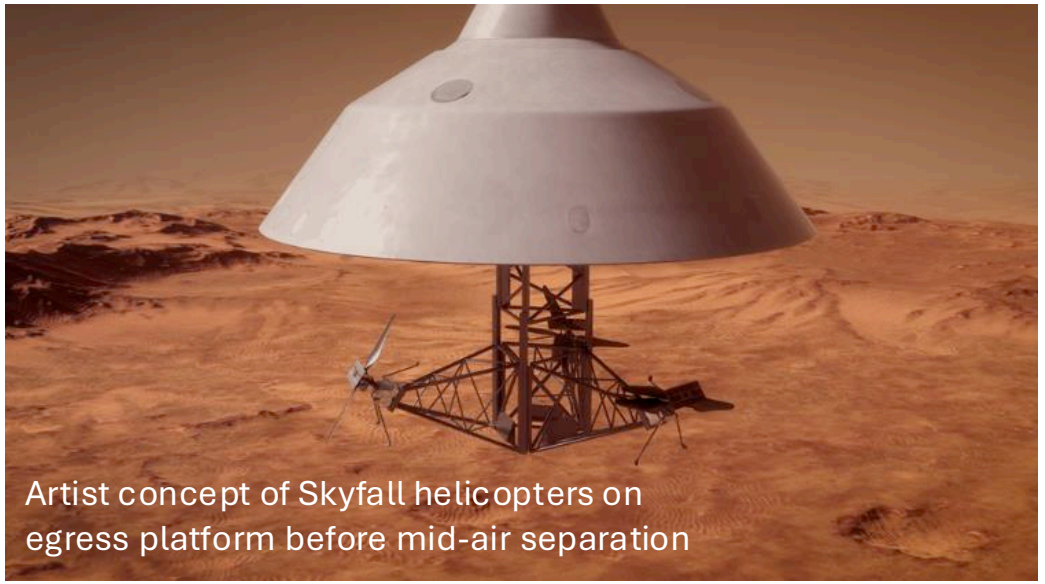
- Preparation for Powered Flight (PPF)
- NFAC PPF: OVERFLOW validation
- Titan PPF

Skyfall Helicopters

- Mission Overview
- Skyfall: overset grids
- Skyfall: hover solution

Mission overview

- Three autonomous Mars helicopters
- Delivered together to the Martian surface
- Designed to explore multiple, geographically separated sites
- Equipped with high-resolution cameras and ground-penetrating radar (**GPR**) to search for subsurface water ice
- Builds on the heritage of **Ingenuity** and extends rotorcraft exploration to a multi-vehicle mission



Artist concept of Skyfall helicopters on egress platform before mid-air separation

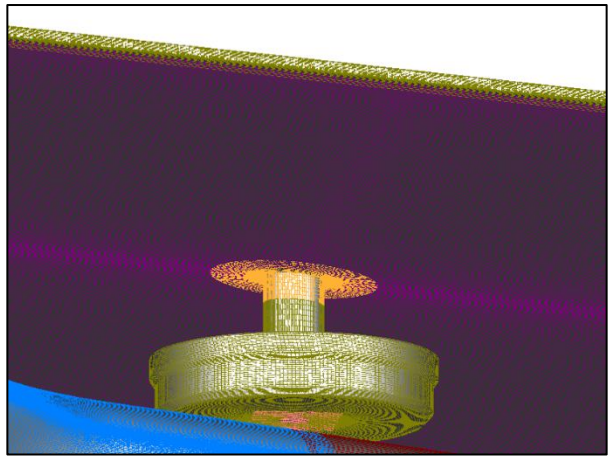
Skyfall helicopter with **GPR** stowed (left) and deployed (right)



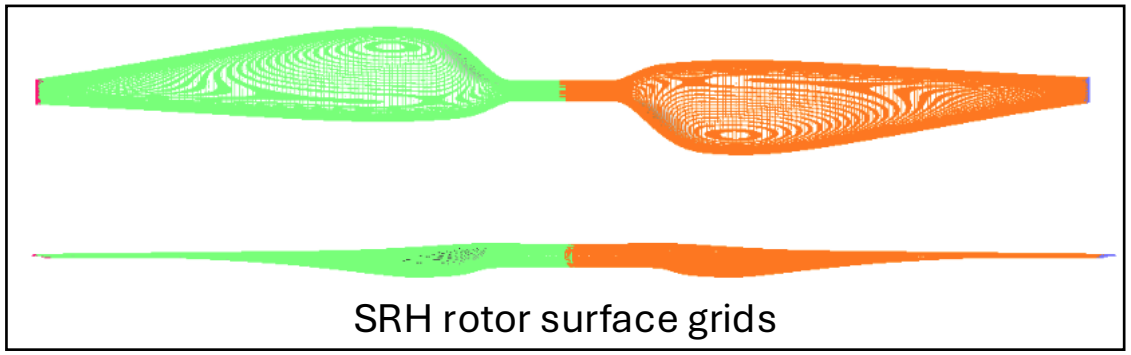


Artist concept of Skyfall helicopters during mid-air separation

Skyfall: overset grid system

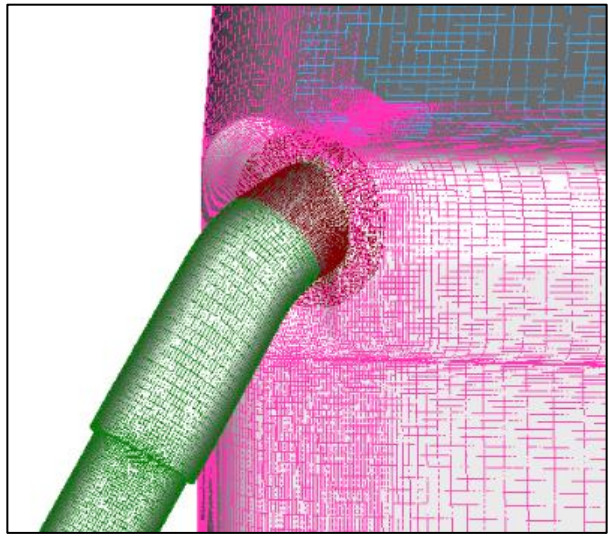


Junction of central axis with solar panel. Solar panel can tilt.

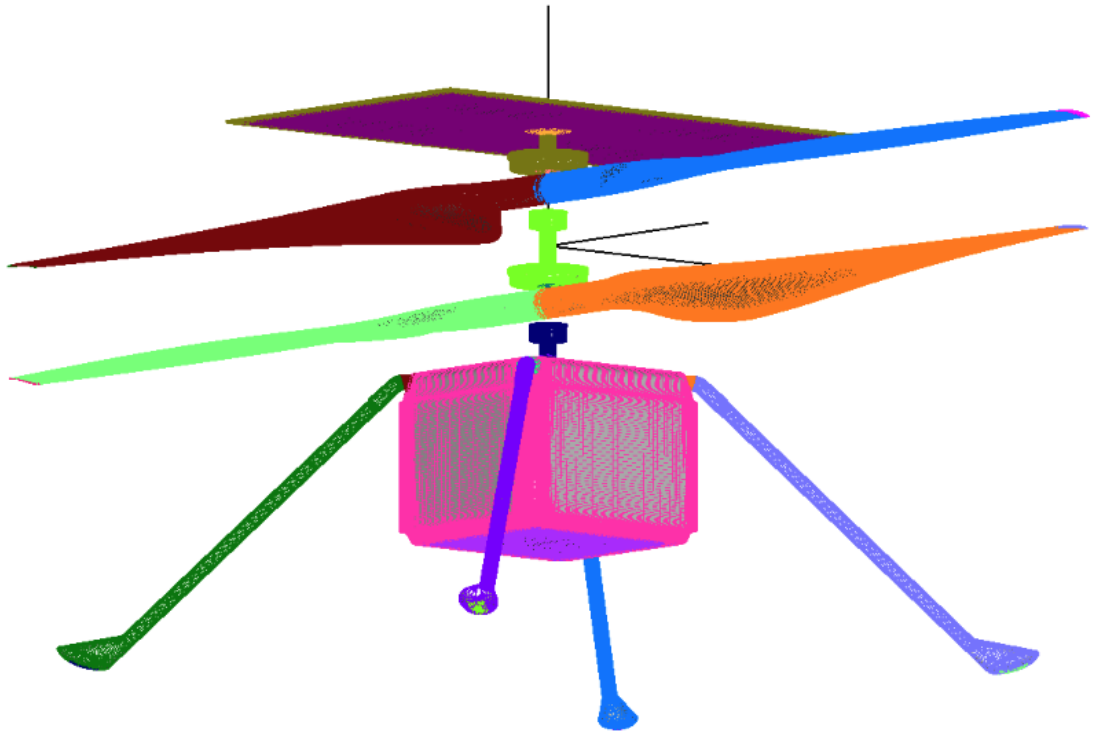


SRH rotor surface grids

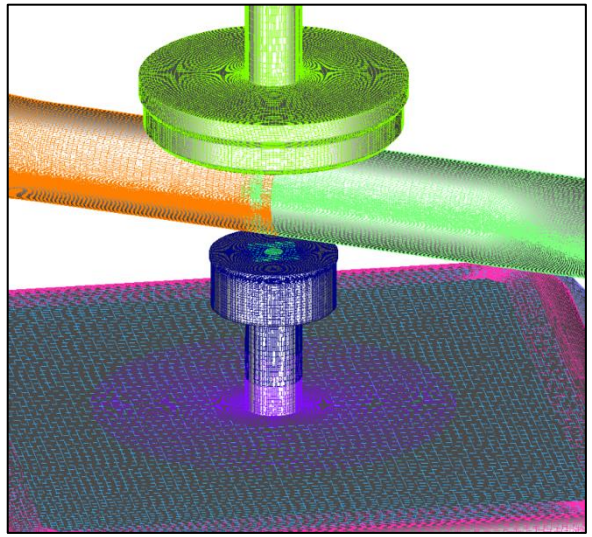
- Grid system contains main components for the Skyfall helicopters, with very few simplifications.
- Initial simulations do not include GPR.
- 35 NB grids with 65 NB million grid points.
- Complete grid system has 230-260 million grid points.



Landing gear-main body grids



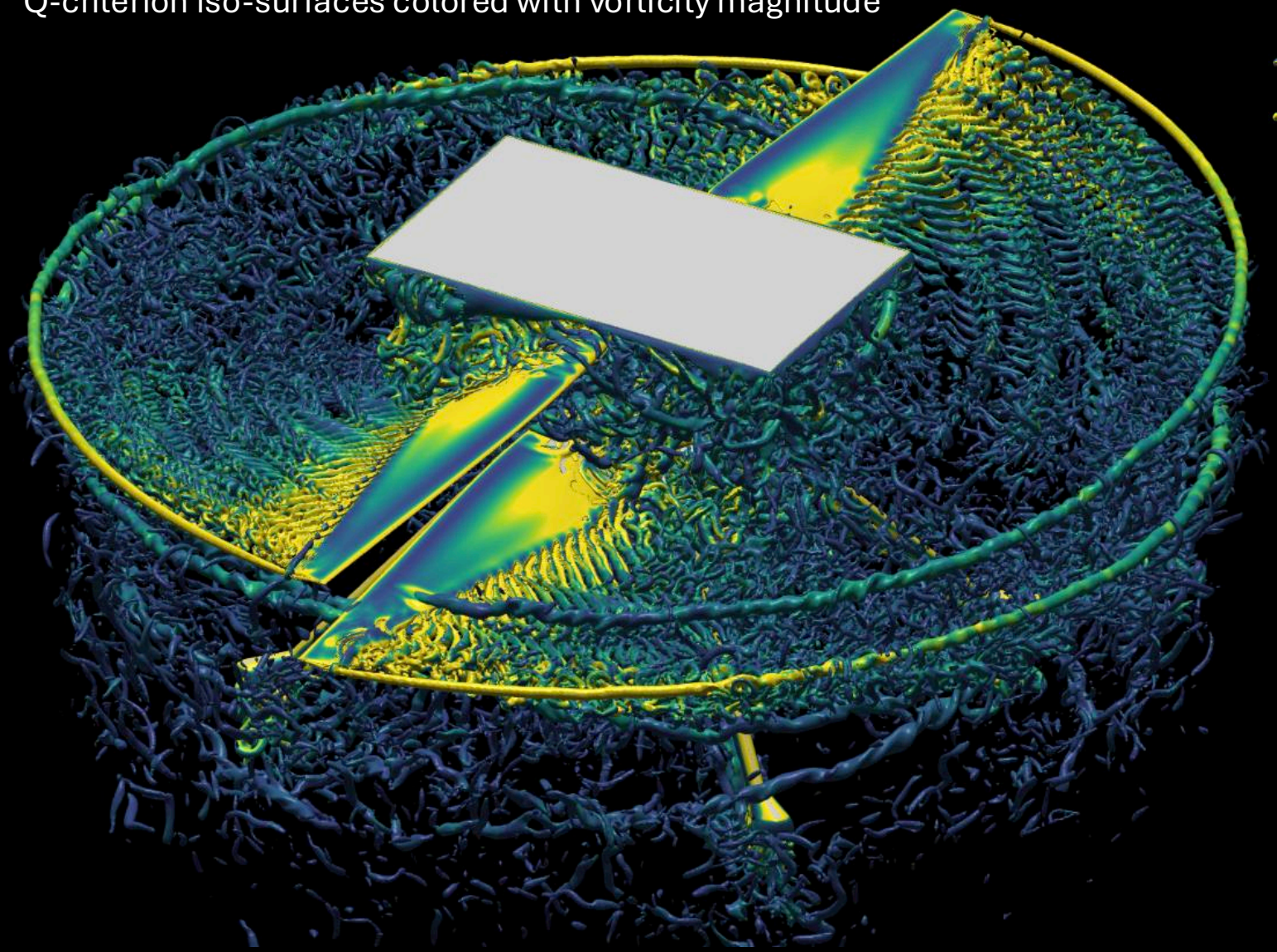
Skyfall complete grid system

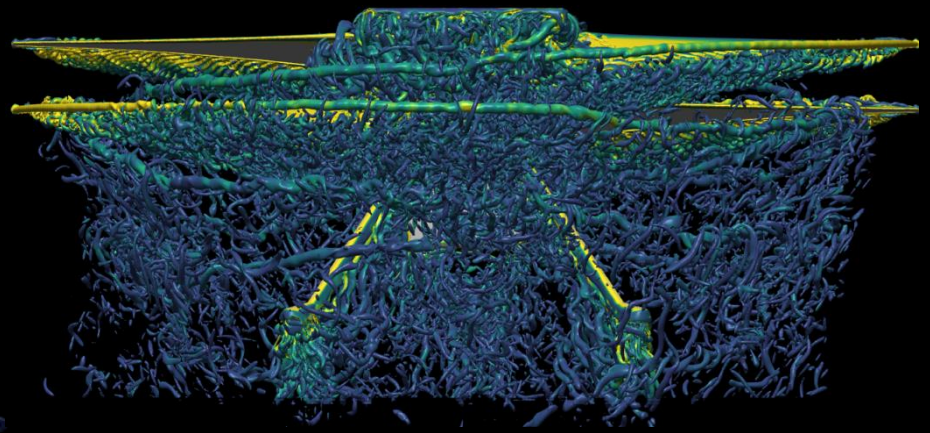


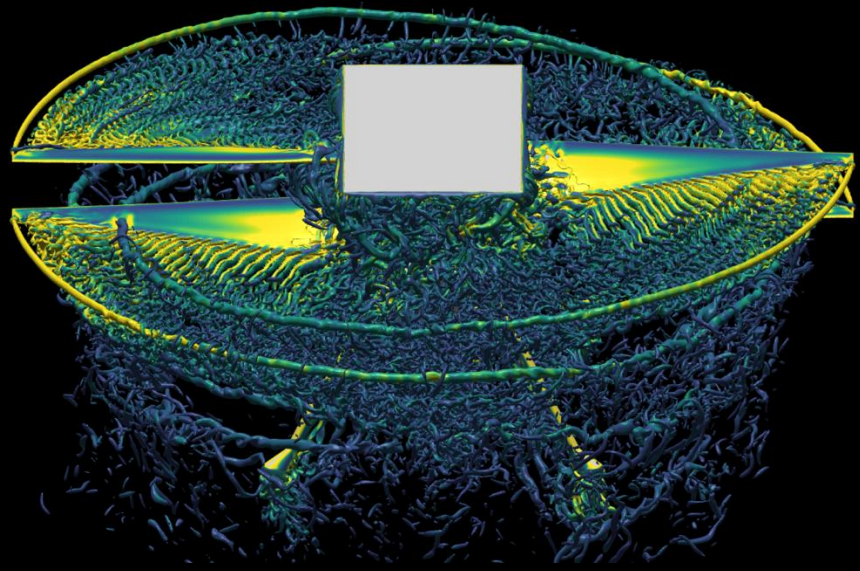
Central axis-main body junction

Skyfall: hover solution

Q-criterion iso-surfaces colored with vorticity magnitude

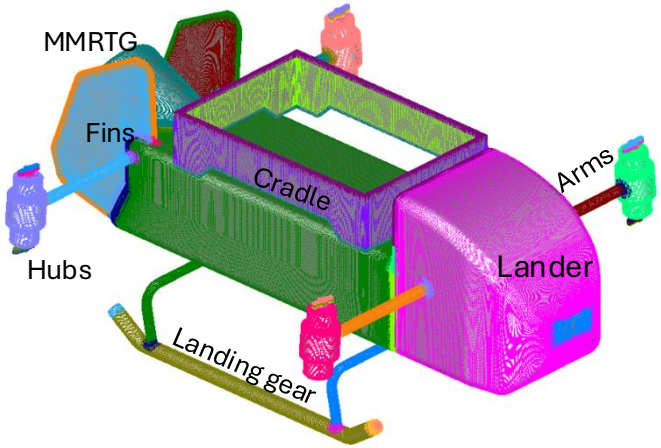






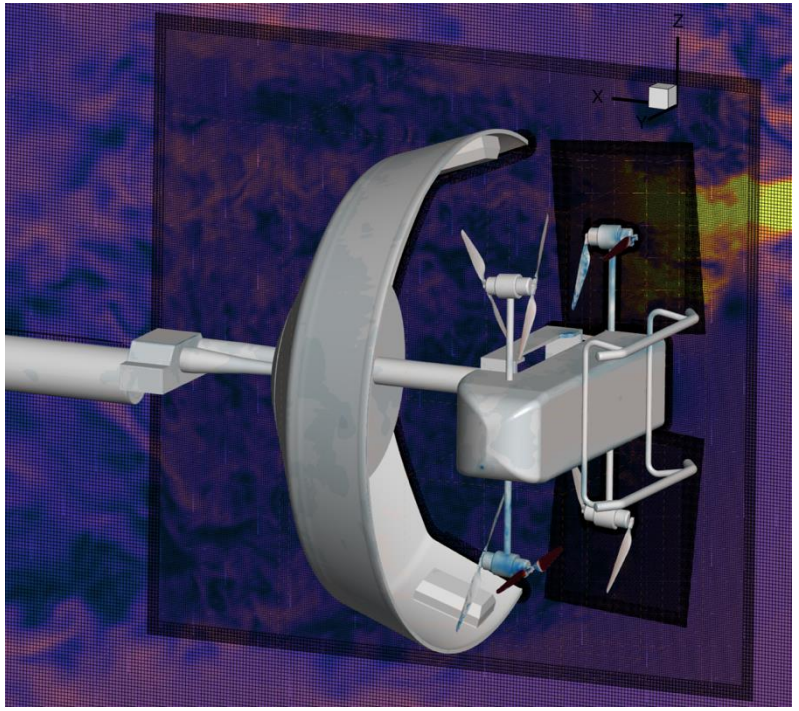
Conclusions

Overset grids enable high-fidelity CFD of complex planetary rotorcraft configurations.



1. **Complex geometries can be decomposed into manageable components**
   Component-based near-body grids were used to represent rotors, vehicle components, backshell, and test hardware.

2. **Grid resolution can be targeted to the relevant flow physics**
   Body-fitted near-body grids resolve the vehicle and blades, while Cartesian off-body grids provide refinement for rotor and backshell wakes.



3. **The methodology scales to very large, complex grid systems**
   - **Dragonfly NFAC:** 228 NB grids | 280 M total points
   - **Dragonfly Titan:** 342 NB grids | 137M NB points | 400–450M total points
   - **Skyfall:** 35 NB grids | 65M NB points | 230–260M total points.

Acknowledgements

Dragonfly

- Dragonfly work supported by the SMD Dragonfly Mission through Johns Hopkins University Applied Physics Laboratory (**APL**).
- Dragonfly aerodynamics & mobility teams. Special thanks to Matt Misiorowski, Michael Marshall, Sven Schmitz, Pat Bowles, and Kate Gruber.

Skyfall

- Skyfall work supported by the Subsonic Vehicle Technologies and Tools (**SVTT**) in collaboration with the Jet Propulsion Laboratory (**JPL**).
- Special thanks to Tove Aagreen, Haley Cummings, and Shannah Withrow.

OVERFLOW & Computational Resources

- OVERFLOW development supported by NASA's former Revolutionary Vertical Lift Technology (**RVLT**) project, now Subsonic Vehicle Technologies and Tools (**SVTT**).
- Computational resources provided by the NASA High-End Computing (**HEC**) Program through the NASA Advanced Supercomputing (NAS) Division at NASA Ames Research Center.






SKYFALL
MARS HELICOPTERS
NASA
ARTIST'S CONCEPT
- MAP RESOURCES
- GROUND PENETRATING RADAR
TO FIND WATER DEPOSITS
JPL
Jet Propulsion Laboratory
California Institute of Technology
NASA
DRAGONFLY
Flights of Exploration Across
Saturn's Moon Titan
APL
JOHNS HOPKINS
APPLIED PHYSICS LABORATORY